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Supplemental information

Accelerated discovery of CO_2-to-C_3-hydrocarbon electrocatalysts with human-in-the-loop

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Methods

Method S1 | Search space definition

The 15-element search space was established based on three key criteria: (i) demonstrated CO_2R activity in previous studies, (ii) economic viability for industrial scaling, and (iii) thermodynamic stability under CO_2R operating conditions. Candidates were first identified from published CO_2R reports (**Table S1**). To ensure scalability, platinum-group metals (PGMs) were excluded based on cost considerations.

A key rationale was to employ adsorption-energy (E_{ads})–based descriptors, which directly encode CO_2R elementary steps with greater mechanistic relevance than composition-only features, such as Magpie. To construct these descriptors efficiently and consistently, we used the AdsorbML pipeline[1,2], which is trained on Open Catalyst (OC) Project datasets of metallic, intermetallic, and oxide surfaces[3,4]. Under electrochemical CO_2R conditions, however, most oxides are rapidly reduced to metallic states, while sulfides and selenides are unstable and are also incompatible with the automated co-reduction synthesis routes in this work. Accordingly, we restricted the element set to those that can persist in metallic phases under CO_2R operating conditions.

We also note that thin surface oxides can form during catalyst preparation steps such as ball milling, ink formulation, or brief ambient-air exposure. Accelerated workflow in this work was not designed to eliminate such pre-catalyst oxidation completely; rather, to ensure that under the operative CO_2R pH–E window, the working surface is metallic, consistent with the descriptor assumptions.

The operating window was determined from COMSOL pH simulations combined with an MEA voltage-loss diagnostic, which indicated a local surface pH of ~12.5 and cathode potentials below −1.6 V_{SHE} at ≥100 mA cm^{-2}. Pourbaix diagrams confirm that elements metallic at −1.6 V_{SHE} remain so at more negative potentials when pH is constant; thus, stability at pH 12.5 and −1.6 V_{SHE} served as the critical selection boundary (**Figure S1**).

Finally, within the Cu-rich regime, we did not consider trace additions of PGMs in this campaign. This decision preserved a closed, feature-fixed element set and mitigated variance inflation under a fixed experimental budget. An “explore-all-then-filter” design—initially including a broader set (including trace PGMs) and applying an ex post economic screen—is a valid alternative, particularly when sample budgets permit.

Method S2 | Acceleration score/factor of experimentation platforms

The acceleration factors of the accelerated experimental platforms were quantified using Acc(X)eleration Performance Indicators (XPIs) framework, previously established for evaluating accelerated materials discovery platforms[5]:

Overall acceleration score = (XPI-1 + XPI-2) × XPI-3 × XPI-4 ÷ XPI-5 (eq 1)

- **XPI-1**: New materials examined
- **XPI-2**: New materials with threshold performance
- **XPI-3**: Performance enhancement of best material
- **XPI-4**: Repeatability and reproducibility
- **XPI-5**: Relative human cost

We adapted this metric system to evaluate (1) the acceleration of experimentation platform, considering throughput and human time-cost, and (2) the acceleration of the overall discovery workflow across various experimental and ML strategies. In this **Method S2**, we will only discuss (1), while (2) will be tackled in **Method S6**. To evaluate the acceleration of the experimentation platform itself, *i.e.*, (1), we adapted XPI-based metric to the following formula:

Acceleration score (AS) = (XPI-1 + XPI-2) × XPI-3 × XPI-4 ÷ XPI-5 (eq 2)

- **XPI-1**: Number of materials synthesized or CO_2R tested
- **XPI-2**: Number of new materials over threshold performance
- **XPI-3**: Performance of best material over time (fold increment)
- **XPI-4**: Repeatability of the experimental platform (precision, %)
- **XPI-5**: Human time-cost to achieve XPI-1 (%, relative to conventional workflow)

Here in eq 2, we excluded the material performance-based indicators (XPI-2 and XPI-3, shown in gray in the equation) to isolate the experimental platform's acceleration capabilities. The acceleration scores (AS) were first evaluated separately for catalyst synthesis and CO_2R performance evaluation processes, with detailed XPI values, and following AS are shown in **Table S2**. We note that these XPI values represent conservative estimates to avoid overreporting the acceleration achieved in the workflow, with detailed rationale for each metric provided below.

<u>Catalyst synthesis</u>

- **XPI-1**: For conventional lab workflow, we assumed that a typical batch of ~6 multi-element catalysts can be synthesized per day. In the accelerated experimental workflow using *Chemspeed* automated synthesis, synthesis of 60 catalysts took approximately 28 hours, resulting in an XPI-1 value of ~51.4 catalysts synthesized per day.

- **XPI-4**: The repeatability assessment of experimental platform. The repeatability score was calculated using the precision indicator:

$$Precision = \left(1 - \frac{\sigma}{\mu}\right) \times 100 \qquad \text{(eq 3)}$$

based on catalyst mass yields from three repeated syntheses of $Cu_{1.00}$ catalyst.

- **XPI-5**: In a conventional lab workflow, synthesis of 6 catalysts requires approximately ~2 hours of human intervention (~10 minutes for stock solution preparation and mixing, 60 minutes for reaction, and ~60 minutes for three rounds of centrifuge particle washing—consisting of three cycles of 10-minute centrifugation and 10-minute redispersion). Although direct human manipulation is not required during the reaction and centrifugation runs, these periods still count toward human cost as they require intermittent interventions—terminating reactions, retrieving samples from the centrifuge, and redispersing catalysts using ultrasonic bath. In contrast, the automated synthesis using *Chemspeed* requires only ~0.5 hours of human involvement for stock solution preparation. Taking the conventional workflow time (~2 hours) as 100%, this translates to an XPI-5 value of ~25% for the automated system.

To sum up, AS for catalyst synthesis,

$$AS_{Synthesis\ (Conventional\ \ lab\ \ procedure)} = 6 \times \frac{94.7\%}{100\%} = \sim 5.7$$

$$AS_{Synthesis\ (Accelerated\ \ experimentation)} = 51.4 \times \frac{94.5\%}{25\%} = \sim 194.3$$

The automated catalyst synthesis demonstrates an acceleration factor of 34.1× over conventional laboratory-based manual synthesis.

<u>CO_2R performance evaluation</u>

- **XPI-1**: For conventional lab workflow, we typically analyze ~ 5 MEAs per day for CO_2R. In contrast, the accelerated CO_2R workflow, utilizing custom MEAs with simple and rapid assembly/disassembly through hold-down clamps, enables analysis of 30 samples per day using a 10× parallelized MEA platform (three rounds of operation). The resulting 90 gas samples (3 current densities × 30 MEA operations) are automatically analyzed within a day, with each analysis taking ~16 minutes (see detailed parallelized MEA-based CO_2 electrolysis in **Method S3**).

- **XPI-4**: The repeatability assessment for both experimental platforms was conducted using $Cu_{1.00}$ as a reference catalyst. The repeatability scores were calculated using the precision indicator calculation above (eq 3) based on the mean and standard deviation of propylene FEs from three repeated measurements.

- **XPI-5**: In a conventional lab workflow, analyzing one catalyst in MEA configuration at three different current densities requires ~1 hour (5 minutes for assembly, three cycles of [3 minutes catalyst activation + 15 minutes GC analysis], and 3 minutes for disassembly). We assumed a typical daily throughput of 5 MEA analyses in 5 hours of lab work. In contrast, the 10× parallelized MEA platform with rapid assembly/disassembly custom MEAs require ~ 42 minutes per operation round (**Figure S8**). With three rounds analyzing 30 samples, the total human intervention time is ~2 hours. Taking the conventional workflow time (~ 5 hours for 5 MEAs) as 100%, this translates to an XPI-5 value of ~33.3% for the accelerated system.

To sum up, AS for CO_2R performance evaluation,

$$AS_{CO_2R\ (Conventional\ \ lab\ \ procedure)} = 6 \times \frac{81.6\%}{100\%} = \sim 4.1$$

$$AS_{CO_2R\ (Accelerated\ \ experimentation)} = 30 \times \frac{82.3\%}{33.3\%} = \sim 74.1$$

Parallelized and automated CO_2R platform demonstrates an acceleration factor of 18.1× over conventional laboratory workflow.

<u>Overall end-to-end experimental workflow</u>

To accurately assess the comprehensive acceleration of the entire experiment workflow, we must consider all experimental components: catalyst synthesis, electrode preparation, CO_2R testing and idle time between processes. For a more realistic evaluation, we calculated the overall acceleration on a weekly basis, accounting for both experimental throughput and corresponding human time-cost.

- **XPI-1**: For conventional lab workflow, we assumed that 6 catalysts could be synthesized and evaluated for CO_2R with 2 rounds per week, resulting in a weekly screening capacity of 12 catalysts. (This represents a highly challenging yet maximally achievable scenario, deliberately established to reflect the maximum possible throughput under conventional lab conditions.) In contrast, the accelerated experimentation workflow enables screening of 60 catalysts per week. While all 60 catalysts can be synthesized in a single automated run, the CO_2R evaluation is performed over two days, with 30 samples per day using a 10× parallelized MEA platform (three rounds of operation).
- **XPI-4**: The overall repeatability of the experimental workflow was calculated as the product of the repeatability values obtained from the catalyst synthesis platform and the CO_2R evaluation platform.
- **XPI-5**: As illustrated in **Figure 10B**, when considering the complete conventional workflow encompassing catalyst synthesis, electrode preparation (including ~1 hour for ink preparation and ~0.5 hours per catalyst for spray coating), the comprehensive screening of 12 catalysts per week would require ~24 hours of human time-cost. In contrast, the accelerated experimentation workflow (**Figure 10D**) requires a human time-cost of only ~4.5 hours including automated synthesis, electrode preparation and parallelized, automated CO_2R evaluation. Taking the conventional workflow time (~24 hours) as 100%, this translates to an XPI-5 value of ~18.8% for the automated system.

To sum up, AS for overall end-to-end experimental workflow,

$$AS_{End-to-End\ (Conventional\ \ lab\ \ procedure)} = 12 \times \frac{77.3\%}{100\%} = \sim 9.3$$

$$AS_{End-to-end\ (Accelerated\ \ experimentation)} = 60 \times \frac{77.8\%}{18.8\%} = \sim 248.2$$

As a result, considering the end-to-end workflow, accelerated experimentation workflow demonstrates an acceleration factor of 26.7× over the conventional laboratory workflow.

Method S3 | Parallelized MEA-based CO_2 electrolysis with automated GC analysis

Parallel MEA-based CO_2 electrolysis

The CO_2 electrolysis experiments were conducted at three current densities (100, 200 and 300 mA cm^{-2}), with each current density maintained for 9 minutes: 3 minutes for catalyst activation, 5 minutes for gas sampling in vial and a minute buffer period (**Figure S8**). The complete evaluation cycle for three current densities required 27 minutes in total, yielding representative voltage profiles (**Figure S9**). Each catalytic composition was evaluated on three randomly selected MEAs from a set of ten MEA cells, with triplicate CO_2R measurements to obtain means and standard deviations. Some datapoints affected by unexpected operational failures, such as anolyte leakage, were excluded from statistical analysis to ensure data reliability (detailed measurement counts and MEA identification numbers for each composition are provided in the dataset).

Gas product sampling during parallel MEA operation

Gas sampling was performed using a dedicated sampling stand connected to ten cathode outlets (**Figure S5A–B**). The process involved plugging in capped headspace vials (21.68 mL volume) for the 5-minute sampling period, followed by vial removal during the 1-minute buffer period. The sampling system utilized dual plugged needles for simultaneous gas sample introduction and air purging (**Figure S5B**). The 5-minute sampling duration was chosen to ensure sufficient purging of gas samples under 30 sccm CO_2 feed conditions. Three rounds of 10x MEA operation generated 90 gas samples across three current densities for 30 catalysts, requiring only 2.1 hours of laboratory work.

Automated gas product analysis of collected gas samples

Gas product analysis was conducted using gas chromatography (GC) with a 14-minute chromatogram acquisition time per sample in a total GC cycle time of 16 minutes, which represents the complete duration between consecutive sample injections, including both analysis and system reset periods. While the process requires 3 minutes of thermostatting (80°C oven) and 1 minute of carrier gas pressurization before each sample injection, these two steps are integrated within the previous sample's GC cycle time, maintaining the 16-minute per-sample duration. This enabled automated analysis of 90 samples in exactly 24 hours.

Data processing

The resulting chromatograms were processed through *PerkinElmer* software's batch processing to obtain GC peak areas automatically. These data were subsequently analyzed using Python code to calculate means and standard deviations of Faradaic efficiencies for gas products at each current density across catalyst compositions. The open-source code for clustering analyses is available on the GitHub repository.

Method S4 | Feature engineering and model refinement for interactive domain knowledge-in-the-loop data acquisition

To interpret the Generalized Additive Model (GAM)-based ML model and refine its shape function based on on-the-fly domain knowledge gleaned from the dataset, we sought to develop descriptors that are intuitively understandable to enable manual editing by human researchers in a human-in-the-loop data acquisition approach.

The descriptor consists of two main types of information: compositional information and CO_2R intermediate adsorption energy (E_{ads})-based information that can encode the CO_2R mechanistic pathway of constituent elements. For this purpose, we engineered the following descriptors for each composition, as described below:

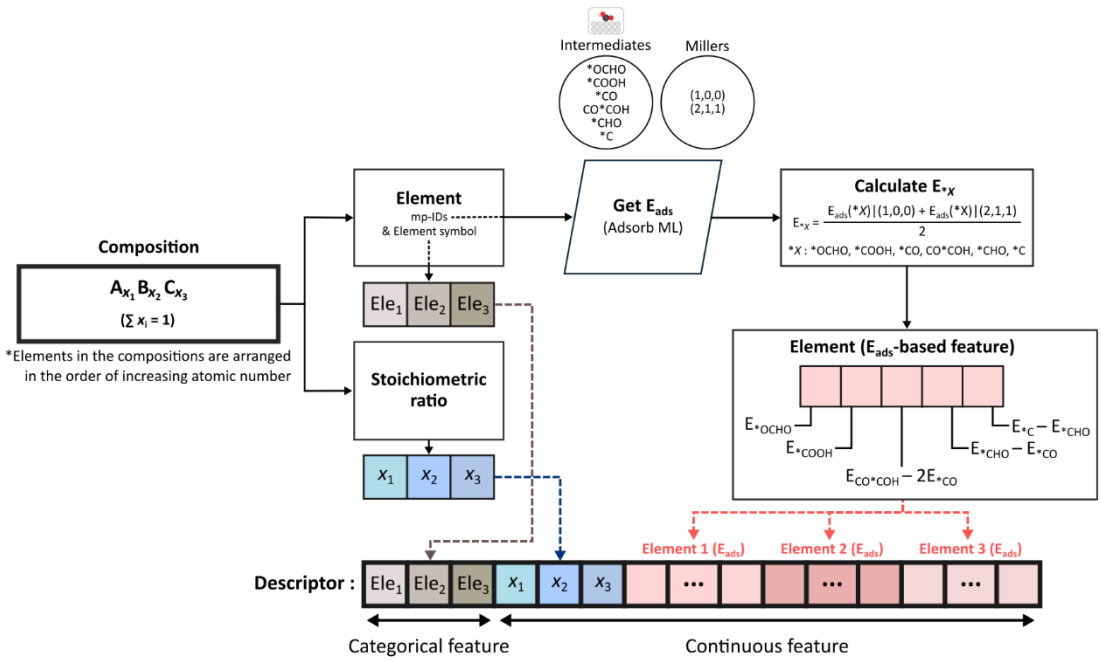

Figure S10. Schematic illustration of feature engineering for descriptor construction in the human-in-the-loop framework.

<u>Compositional information (element symbol-based categorical feature)</u>

To capture compositional information, we implemented an atomic number-ordered representation of each composition and created categorical features labeled as Ele_1, Ele_2 and Ele_3 based on their elemental symbols. Additionally, we incorporated their corresponding stoichiometric ratios, x_1, x_2 and x_3, as descriptors. For unary or binary compositions that lack second or third elements, we populated the missing element fields with NaN values to ensure consistent feature dimensionality across all compositions.

<u>CO_2R intermediate E_{ads}-based continuous features</u>

In electrocatalytic CO_2R, the free energy (G) differences between reaction intermediates are crucial determinants of the thermodynamic driving force. However, full free energy calculations are computationally expensive and complex, making it practically impossible to perform comprehensive calculations for all potential materials, i.e., ~200 million compositions

in 15-element space with 1 at% steps, to use them as descriptors.

E_{ads} of reaction intermediates have been well-established as powerful descriptors that effectively correlate with experimental performance metrics including activity and selectivity, enabling predictive screening of catalysts based on computational insights. E_{ads} calculations, especially related to CO_2R, have recently been accelerated through approaches like AdsorbML, which leverages ML potentials combined with single point DFT calculations to identify low-energy adsorbate-surface configurations[1,3]. This makes E_{ads}-based featurization computationally accessible[6]. For these reasons, we engineered E_{ads}-based descriptors to encode the thermodynamic driving forces of well-established reaction pathways in CO_2R.

We obtained the E_{ads} of key CO_2R intermediates (*X = *OCHO, *COOH, *CO, CO*COH, *CHO and *C, here C* as a representative of *CH_x as E_{ads}(*CH_x) vary linearly with E_{ads}(*C)[7]) on the most stable phases of each element under CO_2R operating conditions (as discussed in **Method S1**). E_{ads} values were determined on two representative Miller index surfaces: (100) and (211), representing low and high Miller indices, respectively[8,9]. These two facets respectively sample terrace and stepped coordination environments—(100) capturing terrace chemistry[8-10] and (211) serving as a canonical stepped surface—providing a compact yet mechanistically meaningful representation with strong precedent in first-principles catalysis[8,11]. Subsequently, we averaged the E_{ads} values from these two Miller indices to derive the E_{*X} descriptor for each adsorbate. It is well known in ML that increasing the number of features reduces performance in the small-data regime. As adsorption energies are known to follow the coordination-number trend (211 > 100 ≈ 111)[12], the average adsorption energy delivers an "effective" adsorption energy that approximates the ensemble average of terraces and steps likely present under in operando conditions. Using these descriptors, we captured the thermodynamic feasibility of five representative branches in the CO_2R mechanism pathway (**E_{*OCHO}**: CO_2 → *OCHO, **E_{*COOH}**: CO_2 → *COOH, **$E_{CO*COH} - 2E_{*CO}$**: 2*CO → CO*COH, **$E_{*CHO} - E_{*CO}$**: *CO → *CHO, **$E_{*C} - E_{*CO}$**: *CO → *C)[13] as shown in **Figure S11** and the figure above. For compositions missing second or third elements, the corresponding E_{ads}-based descriptors were also populated with NaN values. **Table S3** shows all the computational steps involved in the creation of the feature vectors.

Refinement of GAM-based ML model shape functions using on-the-fly domain knowledge

After training the GAM-based ML model, the descriptor detailed above enables intuitive model editing by human researchers.

For features encoding compositional information, when the acquired data suggests that a particular element may positively influence C_3 hydrocarbon formation, researchers can directly assign higher shape function values to that element's categorical feature. However, we refrained from modifying the stoichiometric ratios corresponding to different elements across compositions due to the atomic number ordering—thus direct modification of these values would not provide consistent physical meaning across different compositions; these features serve merely to encode compositional information.

For E_{ads}-based continuous features, negative values indicate thermodynamic feasibility of a

specific reaction pathway, and vice versa. For example, the negative value of $E_{*CHO} - E_{*CO}$ indicates that the E_{ads} of *CHO is lower than that of *CO (stronger binding with *CHO compared to with *CO), suggesting a strong thermodynamic tendency for the conversion from *CO to *CHO. This enables intuitive shape function editing. For instance, as mentioned in the main text, if we observe on-the-fly from the seed dataset that *CO dimerization may correlate with high production of C_3 hydrocarbons, propylene, we can integrate this insight into the model by editing the shape functions corresponding to $E_{CO*COH} - 2E_{*CO}$ for all elements Ele_1–Ele_3, specifically making them decrease stepwise with increasing $E_{CO*COH} - 2E_{*CO}$ values.

Method S5 | Detailed GAM-based ML model refinement and composition sampling for the next iteration using on-the-fly domain knowledge

The detailed shape function edits are documented in **Table S4**. Below, we provide a comprehensive account of how domain knowledge gleaned at each iteration informed corresponding shape function modifications using *GAMChanger*, and how these refined models guided sampling for determining subsequent experimental iterations.

Iteration 1

- **Base Dataset:**
  - Seed dataset (total, 120 datapoints)
- **$j_{propylene}$ Threshold:**
  - 0.05 mA cm^{-2}
- **Observations or Inferred Domain Knowledge from the Base Dataset:** (discussed in the main text)
  - Elements showed notable C_3 hydrocarbon activity:
    - Propylene: Cu + [Pd, Ag, Sn]
    - Propane: [Co, Mn, Fe] + [Ag, Cd, In, Sn]
  - Hypothesis on mechanistic CO_2R pathways:
    Both *CO dimerization and *CO to *CHO conversion may be responsible for C_3 hydrocarbon production
- **Shape Function Modification:**
  - For categorical Ele1, Ele2 features, increase shape function value of Ag, Cd, Co, Cu, Fe, In, Mn, Pd and Sn to 1.5
  - For continuous $E_{CO*COH} - 2E_{*CO}$ and $E_{*CHO} - E_{*CO}$ feature of both Ele1 and Ele2, establish a gradient with 5 equal-size bins where lower feature values were set to 0.5 and higher feature values to – 0.5, creating a stepwise decreasing function—to reflect the hypothesis that more thermodynamically favorable reaction pathways in both directions contribute positively to C_3 hydrocarbon production.
- **Sampling Rationale for the Next Iteration:**
  - Based on the observed differences between Cu-containing and non-Cu compositions, we balanced the 30 samples in Iteration 1 by selecting 15 Cu-containing compositions and 15 non-Cu compositions.
  - For Iteration 1, we focused on acquiring binary compositional data rather than expanding to ternary compositions.
  - To focus on exploring the overall landscape in Iteration 1, we employed distribution-preserving stratified sampling of unexplored compositions, maintaining the True/False prediction ratio from the domain knowledge-aligned refined model. In this context, the True/False ratio refers to the ratio within each targeted sampling category (*e.g.*, non-Cu binary or Cu-containing binary) rather than across the entire compositional space.

- To ensure sufficient compositional diversity among the predicted compositions, we selected samples with 5 at% intervals.

Iteration 2

- **Base Dataset:**
  - Seed dataset + Iteration 1 Data (total 150 datapoints)

- **$j_{propylene}$ Threshold:**
  - 0.1 mA cm^{-2}

- **Observations or Inferred Domain Knowledge from the Base Dataset:**
  - Elements from the compositions that showed over threshold performance: Cu, Pd, Ag, Sn, Fe, Cd and In

Composition	$j_{propylene}$ (mA cm^{-2})
$Cu_{1.00}$	1.629
$Cu_{0.50}Pd_{0.50}$	1.212
$Cu_{0.50}Ag_{0.50}$	1.151
$Cu_{0.50}Sn_{0.50}$	0.414
$Cu_{0.20}Ag_{0.80}$	0.382
$Fe_{0.50}Cd_{0.50}$	0.126
$Fe_{0.75}In_{0.25}$	0.111

- **Supplementary Domain Knowledge:**
  - **Ni** – Although Ni did not exhibit significant C_3 activity in the dataset, we included it based on previous literature report of its activity toward long-chain hydrocarbon production[9].
  - Cu-rich compositions ($x_{Cu} \geq 0.95$) were anticipated to show good propylene performance proportional to their high ethylene production.

- **Shape Function Modification:**
  1. For categorical Ele1, Ele2 features, increase shape function value of Ag, Cd, Cu, Fe, In, **Ni**, Pd and Sn to 1.

- **Sampling Rationale for the Next Iteration:**
  - In Iteration 2, as with Iteration 1, we maintained an equal distribution by selecting 15 Cu-containing and 15 non-Cu compositions among the 30 samples.
  - From the 15 Cu-containing compositions, we deliberately selected 5 Cu-rich compositions ($x_{Cu} \geq 0.95$) based on supplementary domain knowledge. These compositions were formed by combining Cu with five different secondary elements: four that we considered promising and Ni which we were interested in investigating further with atomic ratios randomly selected while maintaining $x_{Cu} \geq 0.95$.
  - For the remaining samples (excluding the 5 deliberately selected Cu-rich compositions), we continued to focus on exploring the overall landscape in Iteration 2 by employing distribution-preserving stratified sampling of unexplored compositions, maintaining the True/False prediction ratio from the domain

knowledge-aligned refined model. Again, in this context, the True/False ratio refers to the ratio within each targeted sampling category (*e.g.*, Cu-rich binary or non-Cu compositions) rather than across the entire compositional space.

- To ensure sufficient compositional diversity among the predicted compositions (except for the 5 deliberately selected Cu-rich compositions), we selected samples with 5 at% intervals.
- For Iteration 2, we still focused on acquiring binary compositional data rather than expanding to ternary compositions.

<u>Iteration 3</u>

- **Base Dataset:**
  - Seed dataset + Iteration 1, 2 Data (total 180 datapoints)
- **$j_{propylene}$ Threshold:**
  - 0.2 mA cm^{-2}
- **Observations or Inferred Domain Knowledge from the Base Dataset:**
  - Elements from the compositions that showed $j_{propylene}$ better than Cu: Cu, Ag and Pd consistently show good propylene performance, while Cd and Ni emerged as new heteroatom dopants that enhanced performance beyond $Cu_{1.00}$ when incorporated into Cu.
- **Supplementary Domain Knowledge:**
  - Ni, which we deliberately incorporated in Iteration 2, showed promising results ($Ni_{0.01}Cu_{0.99}$, $j_{propylene}$ = 0.812 mA cm^{-2}) and Fe was consistently observed to be effective for C_3 hydrocarbon production in non-Cu category, particularly toward propane. Based on this observation we hypothesized that other middle 3*d* transition metals (Co and Mn) incorporated into Cu would show promising activity.
  - Consistently high-performing compositional elements include Cd, In and Sn — elements that are associated with selectivity toward formate[14]. In this context, we considered Pb—which is also good formate catalyst[15]—would also be valuable to screen.
  - Since the target was CO_2 to propylene conversion (a higher-value chemical than propane, as noted in the main text), and the Cu-containing *CO dimerization pathway appeared crucial for this purpose, we decided to increase the importance of *CO dimerization while decreasing the significance of *CO to *CHO conversion to focus specifically on propylene production.
- **Shape Function Modification:**
  - For categorical Ele1, Ele2 features, increase shape function value of Ag, Cd, **Co**, Cu, **Mn**, Ni, Pd, **Pb** and Sn to 1.
  - For continuous $E_{CO*COH} - 2E_{*CO}$ feature of both Ele1 and Ele2, we establish a gradient with 5 equal-size bins where lowest feature values were set to 0.5 and highest feature values to –0.5, creating a stepwise decreasing function—reflecting

the expectation that favorable *CO dimerization would enhance $j_{propylene}$ values.

- For continuous $E_{*CHO} - E_{*CO}$ feature of both Ele1 and Ele2, we set the opposite trend with the lowest values at –0.5 and highest values at 0.5, creating a stepwise increasing function across 5 equal-size bins. Note that in **Iteration 1** we edited both *CO dimerization and *CO to *CHO conversion pathways with the same trend (stepwise decreasing function) based on both of their contributions to C_3 product formation. However, results through Iteration 2 clearly indicated that Cu presence—essential for achieving high $j_{propylene}$ values, likely promoting the *CO dimerization pathway, likely promoting the *CO dimerization pathway. With propylene production as the primary target due to its higher commercial value, we adjusted the model to emphasize this pathway.

- **Sampling Rationale for the Next Iteration:**
  - For Iteration 3, we increased the proportion of Cu-containing compositions to 20 samples, with the remaining 10 samples selected from non-Cu compositions.
  - Among 20 Cu-containing compositions, we exclusively selected Cu-rich compositions ($x_{Cu} \geq 0.95$) to target catalysts with higher $j_{propylene}$ performance and introduced ternary compositions for the first time in 10 of these samples.
  - We continued to focus on exploring the overall landscape in Iteration 3 by employing distribution-preserving stratified sampling of unexplored compositions, maintaining the True/False prediction ratio from the domain knowledge-aligned refined model. In this context, the True/False ratio refers to the ratio within each targeted sampling category (*e.g.*, Cu-rich binary or non-Cu compositions) rather than across the entire compositional space.
  - To ensure sufficient compositional diversity among the predicted compositions, for non-Cu candidates we selected samples with 5 at% intervals.

<u>Iteration 4</u>

- **Base Dataset:**
  - Seed dataset + Iteration 1, 2, 3 Data (total 210 datapoints)

- **$j_{propylene}$ Threshold:**
  - 0.5 mA cm^{-2}

- **Observations or Inferred Domain Knowledge from the Base Dataset:**
  - Elements from the top ten compositions that show high $j_{propylene}$ performance: Cu, Ag, Pd, Cd, Fe and In (with Fe appearing among the high-performing compositions for the first time. Additionally, $Cu_{0.95}In_{0.05}$ with small amounts of In showed promising performance).

Compositions	$j_{propylene}$ (mA cm^{-2})
$Fe_{0.01}Cu_{0.97}Ag_{0.02}$	2.042
$Cu_{0.97}Ag_{0.03}$	1.787
$Cu_{0.99}Cd_{0.01}$	1.736

$Cu_{0.98}Pd_{0.02}$	1.630
$Cu_{1.00}$	1.629
$Cu_{0.97}Pd_{0.01}Ag_{0.02}$	1.450
$Cu_{0.96}Ag_{0.04}$	1.342
$Cu_{0.95}In_{0.05}$	1.285
$Cu_{0.50}Pd_{0.50}$	1.212
$Cu_{0.50}Ag_{0.50}$	1.151

- **Supplementary Domain Knowledge:**
  - Having observed the performance enhancement effect of small amounts of Fe added to Cu in $Fe_{0.01}Cu_{0.97}Ag_{0.02}$, we sought to further investigate the effects of other 3*d* middle transition metals (Mn, Co and Ni). Since we had already modified the shape functions for Co and Mn in the previous iteration (**Iteration 3**), we aimed to verify their effects through sampling rather than additional shape function edits (see below).

- **Shape Function Modification:**
  - For categorical Ele1, Ele2 features, increase shape function value of Ag, Cd, Cu, Fe, **In** and Pd to 2, and for Ele3, increase shape function value of Ag, Cd, Cu, **In** and Pd to 2 because Fe is not present in Ele3 (note that Ele1~3 are ordered by atomic number).
  - For continuous $E_{CO*COH} - 2E_{*CO}$ feature of Ele3—which newly started to contribute because we started to add ternary compositions from Iteration 3—we established a gradient with 5 equal-size bins where lowest feature values were set to 0.5 and highest feature values to –0.5, creating a stepwise decreasing function—reflecting the expectation that favorable *CO dimerization would enhance $j_{propylene}$ values.
  - For continuous $E_{*CHO} - E_{*CO}$ feature of Ele3, we set the opposite trend with the lowest values at –0.5 and highest values at 0.5, creating a stepwise increasing function across 5 equal-size bins (for the same reasons as in the previous iteration).

- **Sampling Rationale for the Next Iteration:**
  - For Iteration 4, we maintained an increased proportion of Cu-containing compositions with 20 samples versus 10 samples of non-Cu compositions. All Cu-containing compositions were Cu-rich ($x_{Cu} \geq 0.95$) to maximize the likelihood of high propylene production, consistent with the previous iteration.
  - To gain more information about Cu-containing ternary compositions, we designated 15 of the 20 Cu-containing compositions as ternary compositions.
  - From Iteration 4, we shifted toward exploitation-focused sampling to specifically target high propylene performance. This approach involved selecting a higher proportion of compositions predicted to be "True (predicted $j_{propylene}$ over threshold)" (sampling with a T/F ratio of 3:2) rather than maintaining the True/False prediction ratio from the refined model prediction. For Cu-rich binary compositions, despite the model predicting 325 True and 575 False compositions, we deliberately sampled more from the True category (9 samples from predicted True compositions versus 6 samples from predicted False compositions). We still included some predicted False compositions to ensure the resulting dataset would retain information about the broader compositional landscape, supporting more comprehensive ML-assisted data

interpretation in subsequent analyses. Again, in this context, the True/False ratio refers to the ratio within each targeted sampling category (*e.g.*, Cu-rich binary or non-Cu compositions) rather than across the entire compositional space.

- For non-Cu candidates to ensure sufficient compositional diversity among the predicted compositions, we selected samples with 5 at% intervals.

Iteration 5

- **Base Dataset:**
  - Seed dataset + Iteration 1, 2, 3, 4 Data (total 240 datapoints)

- **$j_{propylene}$ Threshold:**
  - 1.0 mA cm^{-2}

- **Observations or Inferred Domain Knowledge from the Base Dataset:**
  - Elements from the top five compositions that show high $j_{propylene}$ performance: Cu, Ag, Pd, Cd, Fe and In. We confirmed the effectiveness of In, which was added based on the previous domain knowledge.

Compositions	$j_{propylene}$ (mA cm^{-2})
$Cu_{0.97}Pd_{0.01}In_{0.02}$	2.262
$Fe_{0.01}Cu_{0.97}Ag_{0.02}$	2.042
$Cu_{0.97}In_{0.03}$	1.885
$Cu_{0.97}Ag_{0.03}$	1.787
$Cu_{0.99}Cd_{0.01}$	1.736

- **Supplementary Domain Knowledge:**
  - No additional supplementary inputs beyond information about high-performing elements.

- **Shape Function Modification:**
  - For categorical Ele1, Ele2 features, increase shape function value of Ag, Cd, Cu, Fe, In and Pd to 2, and for Ele3, increase shape function value of Ag, Cd, Cu, In and Pd to 2 because Fe is not present in Ele3 (Note again that Ele1~3 are ordered by atomic number).
  - We decided against further editing of adsorption energy-based continuous features.

- **Sampling Rationale for the Next Iteration:**
  - In Iteration 5, aiming to aggressively target high propylene production performance through exploitation-focused sampling, we increased the proportion of Cu-rich compositions to 25 out of 30 samples, with 15 ternary Cu-rich and 10 binary Cu-rich compositions to expand the knowledge of ternary systems.
  - As in the previous iteration, we selected a higher proportion of compositions predicted to be "True (predicted $j_{propylene}$ over threshold)" rather than maintaining the True/False prediction ratio from the refined model prediction. However, we implemented a significantly higher T:F ratio (4:1) compared to Iteration 4.

<u>Iteration 6</u> (Additional campaign to collect 300 datapoints in total)

- **Base Dataset:**
  - Seed dataset + Iteration 1, 2, 3, 4, 5 Data (total 270 datapoints)

- **$j_{propylene}$ Threshold:**
  - 1.0 mA cm^{-2}

- **Shape Function Modification & Sampling Rationale for the Next Iteration:**
  - For Iteration 6, which was primarily designed to reach the target of 300 datapoints in total, we followed the same procedure as in the previous iteration used for Iteration 5 selection, but without specifically targeting the T/F ratio. This approach aimed to maximize information collection about the overall landscape.

Method S6 | Acceleration score/factor of discovery workflows across various experimental and ML strategies

We again adapted and applied the XPI-based acceleration evaluation metric to assess the overall acceleration score/factor of workflow regarding "catalyst discovery" aspect across various experimental and ML strategies. For this acceleration evaluation, we included XPI-3 as follows:

$$\textbf{Acceleration score (AS)} = (\textbf{XPI-1} + \textbf{XPI-2}) \times \textbf{XPI-3} \times \textbf{XPI-4} \div \textbf{XPI-5} \qquad \text{(eq 4)}$$

- **XPI-1**: Number of catalysts evaluated per week
- **XPI-2**: Number of new materials with threshold performance
- **XPI-3**: Performance of best material over time (fold increment)
- **XPI-4**: Repeatability of the experimental platform (precision, %)
- **XPI-5**: Human time-cost to achieve threshold performance (%, relative to conventional workflow, here 'BO with batch size = 1 strategy')

The AS assessment required consideration of:

1. Simulated experiments on a virtual experimental dataset (**Figure 3B** in the main text and **Figure S16** with results of BO batch size = 1)
2. Weekly data throughput (**Figure S10**).
3. Human time-cost comparison between conventional laboratory procedure and accelerated experimentation workflows (**Figure S10**).

For the "catalyst discovery" aspect, the reference point for acceleration score calculation was set to when a catalyst with propylene partial current density ($j_{propylene}$) exceeding the threshold of 3.00 mA cm^{-2} was first discovered. Accordingly, **XPI-3** represents the folds of improvement achieved relative to the best performance at the starting point (*i.e.*, seed dataset where the highest $j_{propylene}$ performance was 1.63 mA cm^{-2} from $Cu_{1.00}$ catalyst) when first exceeding the threshold. **XPI-5** represents the human time-cost required to achieve this threshold performance. All XPI-5 values were calculated as percentages relative to the human time-cost of 'Bayesian optimization (BO) batch size = 1 strategy'.
Detailed XPI values and following AS results are shown in **Table S5**. We note again that these XPI values are conservative estimates to avoid overestimating the workflow's acceleration performance. Detailed rationale for each metric is provided below.

<u>BO (batch size = 1, *i.e.*, conventional laboratory procedure)</u>

- **XPI-1**: For conventional experimentation with batch size of 1, we assume that 2 rounds of end-to-end experiment (synthesis → electrode preparation → CO_2R evaluation) can be performed per week (**Figure S10A**), resulting in a total of <u>2 catalysts</u> evaluated weekly.
- **XPI-3**: Performance of 3.07 mA cm^{-2} was achieved in the 168th experiment (**Figure S16**). This represents a <u>1.88-fold</u> improvement relative to the best performance in the seed dataset (*i.e.*, $Cu_{1.00}$).

- **XPI-4**: The overall repeatability of the experimental workflow was calculated as the product of the repeatability values obtained from the catalyst synthesis platform and the CO_2R evaluation platform. Following conventional synthesis and MEA evaluation methods, this value is 77.3% detailed in **Table S2**.
- **XPI-5**: With a weekly throughput of 2 catalysts, reaching the 168th experimental round requires ~84 weeks, resulting in a total human time-cost of 672 hours (84 weeks × 8 hours per week). All subsequent XPI-5 evaluations are presented as percentages relative to this 672-hour human time-cost (considered as 100%) to determine the overall acceleration score.

Therefore,

$$AS_{(BO,\ \ Batch\ size=1)} = 2 \times 1.88 \times \frac{77.3\%}{100\%} = \sim 2.9$$

BO (batch size = 6, i.e., conventional laboratory procedure)

This approach most closely approximates the traditional workflow commonly performed in laboratories.

- **XPI-1**: For conventional experimentation with batch size of 6, we assume that 2 rounds of end-to-end experiment can be performed per week (**Figure S10B**), resulting in a total of 12 catalysts evaluated weekly.
- **XPI-3**: Performance of 3.00 mA cm^{-2} was achieved in the 408th experiment (**Figure S16**). This represents a 1.84-fold improvement relative to the best performance in the seed dataset (*i.e.*, $Cu_{1.00}$).
- **XPI-4**: The overall repeatability of the experimental workflow was calculated as the product of the repeatability values obtained from the catalyst synthesis platform and the CO_2R evaluation platform. Following conventional synthesis and MEA evaluation methods, this value is 77.3% as detailed in **Table S2**.
- **XPI-5**: With a weekly throughput of 12 catalysts, reaching the 408th datapoint requires ~34 weeks, resulting in a total human time-cost of 816 hours (34 weeks × 24 hours per week)—corresponds to 121.4% relative to that of 'BO with batch size = 1 strategy'.

Therefore,

$$AS_{(BO,\ \ Batch\ size=6)} = 12 \times 1.84 \times \frac{77.3\%}{121.4\%} = \sim 14.1$$

BO (batch size = 30, *i.e.*, accelerated experimentation)

- **XPI-1**: For accelerated experimentation workflow with batch size of 30, we assume that 30 catalysts can be evaluated weekly (**Figure S10C**).
- **XPI-3**: Performance of 3.20 mA cm^{-2} was achieved in the 900th experiment (**Figure S16**). This represents a 1.97-fold improvement relative to the best performance in the seed dataset (*i.e.*, $Cu_{1.00}$).

- **XPI-4**: The overall repeatability of the experimental workflow was calculated as the product of the repeatability values obtained from the catalyst synthesis platform and the CO_2R evaluation platform. Following automated synthesis, parallelized MEA operation and automated gas product analysis, this value is 77.8% as detailed in **Method S2**.
- **XPI-5**: With a weekly throughput of 12 catalysts, reaching the 900th datapoint requires ~26 weeks, resulting in a total human time-cost of 65 hours (26 weeks × 2.5 hours per week)—corresponds to 9.7% relative to that of 'BO with batch size = 1 strategy'.

Therefore,

$$AS_{(BO,\ \ Batch\ size=30)} = 30 \times 1.97 \times \frac{77.8\%}{9.7\%} = \sim 474.0$$

Domain knowledge-in-the-loop (batch size = 30, i.e., accelerated exploration)

This methodology represents the actual implementation of accelerated exploration.

- **XPI-1**: For accelerated experimentation workflow with batch size of 30, we assume that 30 catalysts can be evaluated weekly (**Figure S10C**).
- **XPI-3**: Performance of 3.00 mA cm^{-2} was achieved in the 270th experiment (**Figure S16**). This represents a 1.84-fold improvement relative to the best performance in the seed dataset (*i.e.*, $Cu_{1.00}$).
- **XPI-4**: The overall repeatability of the experimental workflow was calculated as the product of the repeatability values obtained from the catalyst synthesis platform and the CO_2R evaluation platform. Following automated synthesis, parallelized MEA operation and automated gas product analysis, this value is 77.8% as detailed in **Method S2**.
- **XPI-5**: With a weekly throughput of 12 catalysts, reaching the 270th datapoint requires ~5 weeks, resulting in a total human time-cost of 12.5 hours (5 weeks × 2.5 hours per week)—corresponds to 1.9% relative to that of 'BO with batch size = 1 strategy'.

Therefore,

$$AS_{(Domain\ knowledge-in-the-loop,\ \ Batch\ size=30)} = 30 \times 1.84 \times \frac{77.8\%}{1.9\%} = \sim 2314.0$$

Method S7 | ML-assisted analyses using adsorption energy descriptors

Using ML-assisted analyses, we aimed to identify potential phases that could form from the pre-catalysts under CO_2R operating conditions. To this end, we sought to develop descriptors by incorporating adsorption energies (E_{ads}) of key CO_2R intermediates on the surfaces of these potential phases to effectively featurize each catalyst composition in an interpretable fashion. Detailed featurization method is illustrated below.

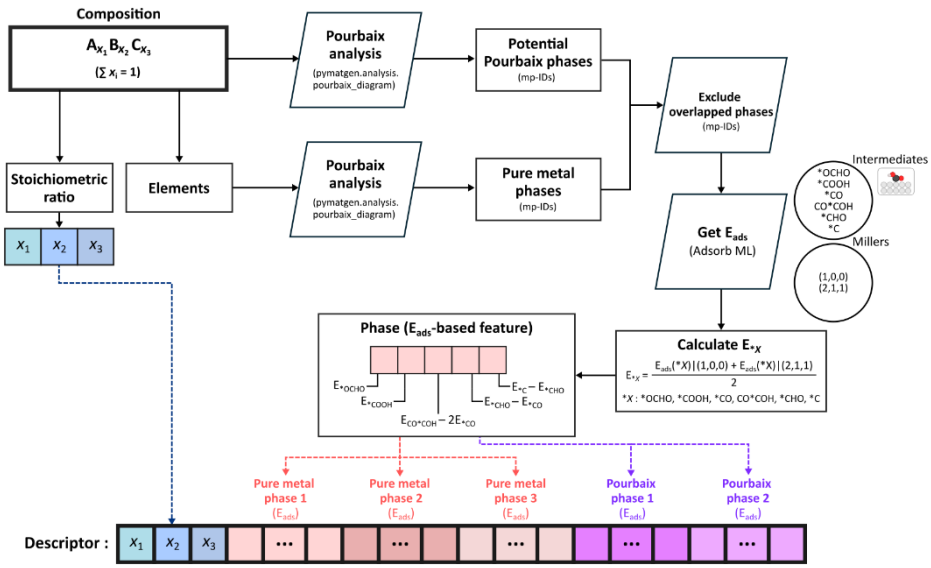

Figure S20. Schematic illustration of featurization method using adsorption energy–based descriptors derived from Pourbaix phase analysis.

The potential phases from a specific composition would include: (1) pure metal phases of constituent elements and (2) phases formed from combinations of constituent elements—which are Pourbaix-stable under CO_2R operating conditions. To identify these phases, we conducted Pourbaix analysis (using pymatgen.analysis.pourbaix_diagram module) on both the multi-component composition itself and its constituent elements to determine potential phases under CO_2R conditions (see **Method S1** for detailed condition) and obtained their corresponding Materials Project IDs (mp-IDs).

After identifying the potential phases, we excluded any overlapping phases (*i.e.*, mp-IDs) to avoid redundancy in the descriptor set (*i.e.*, if any phase from (2) overlapped with pure element phases from (1), we removed it from the Pourbaix phases). We then derived the E_{*X} and constructed descriptors that could capture the thermodynamic feasibility of five representative branches by following the flow described in **Method S4**. To these, we added the stoichiometric ratios, x_1, x_2 and x_3, of each element to complete the descriptors as shown in the figure above. For compositions where certain pure element phases 1–3 or Pourbaix phases 1–2 did not exist, we populated these missing values with NaNs to ensure consistent feature dimensionality across all compositions.

Using these engineered descriptors, we trained XGBoost-based regression models on the 300-composition dataset to predict partial current densities for propylene ($j_{propylene}$) and propane ($j_{propane}$). To rigorously evaluate model performance while maximizing utilization of the limited

dataset, we employed leave-one-out cross-validation (LOOCV). This approach involved training the model on ($n - 1$) samples and evaluating on the remaining sample, repeated across all 300 compositions. Concurrently, we performed SHapley Additive exPlanations (SHAP) analysis on each test sample during the LOOCV process, aggregating these SHAP values across all 300 data points to quantify feature importance and elucidate mechanistic insights from the trained models.

Method S8 | Structural characterization of $Cu_{0.98}In_{0.02}$ catalyst

The combined results of comprehensive characterization using XRD with Rietveld refinement and XRF analysis demonstrate that the catalyst structure consists primarily of dilute In incorporation into the Cu matrix in a solid solution form, with trace amounts of indium present as $In(OH)_3$.

Characterization Results

- **XRD pattern**: The XRD pattern shows characteristic Cu, Cu_2O, and $In(OH)_3$ peaks with no discrete In or In_2O_3 peaks.
- **Rietveld refinement of XRD pattern (Figure S24)**: Cu (metallic): 98.4 wt%, Cu_2O: 1.3 wt%, In(OH)3: 0.3 wt%
- **XRF**: Cu 98.5 at%, In 1.5 at%

Quantitative analysis of In distribution

Based on the bulk analysis from XRD Rietveld refinement, the atomic ratio of Cu and In species can be calculated as follows:

- Cu (total) : $\frac{\frac{98.4\ wt\%}{63.55\frac{g}{mol}}+\frac{1.3\ wt\%}{143.1\frac{g}{mol}}}{\frac{98.4\ wt\%}{63.55\frac{g}{mol}}+\frac{1.3\ wt\%}{143.1\frac{g}{mol}}+\frac{0.3\ wt\%}{165.84\frac{g}{mol}}} \times 100 = 99.87\ \%$
- In (in $In(OH)_3$) : $\frac{\frac{0.3\ wt\%}{165.84\frac{g}{mol}}}{\frac{98.4\ wt\%}{63.55\frac{g}{mol}}+\frac{1.3\ wt\%}{143.1\frac{g}{mol}}+\frac{0.3\ wt\%}{165.84\frac{g}{mol}}} \times 100 = 0.12\ \%$

Therefore, Cu(total) : In (in $In(OH)_3$) = 99.87 : 0.12

From XRF analysis: Cu(total) : In (in $In(OH)_3$) + In (in Cu matrix) = 98.5 : 1.5

- Setting Cu(total) = 98.5 at% as the reference:
- In (in $In(OH)_3$) = 98.5 × (0.12/99.87) = 0.118 at%
  In (in Cu matrix) = 1.5 − 0.118 = 1.382 at%

Therefore: **In (in $In(OH)_3$) : In (in Cu matrix) = 0.118 : 1.382 = 7.9 % : 92.1 %**

Structural interpretation

This quantitative analysis reveals that approximately 92.1 % of In exists as solid solution in the Cu matrix, and only 7.9 % exists as the $In(OH)_3$ phase, which supports the conclusion that "dilute In incorporation in Cu solid solution is the major structural form". TEM/EDX analysis in **Figure 5A** further confirms that In is distributed mostly uniformly throughout the Cu matrix, consistent with solid solution formation. The dilute In incorporation (~1.4 at% in the Cu matrix) is well within the thermodynamic solubility limit of In in Cu[16], ensuring structural stability.

Figures

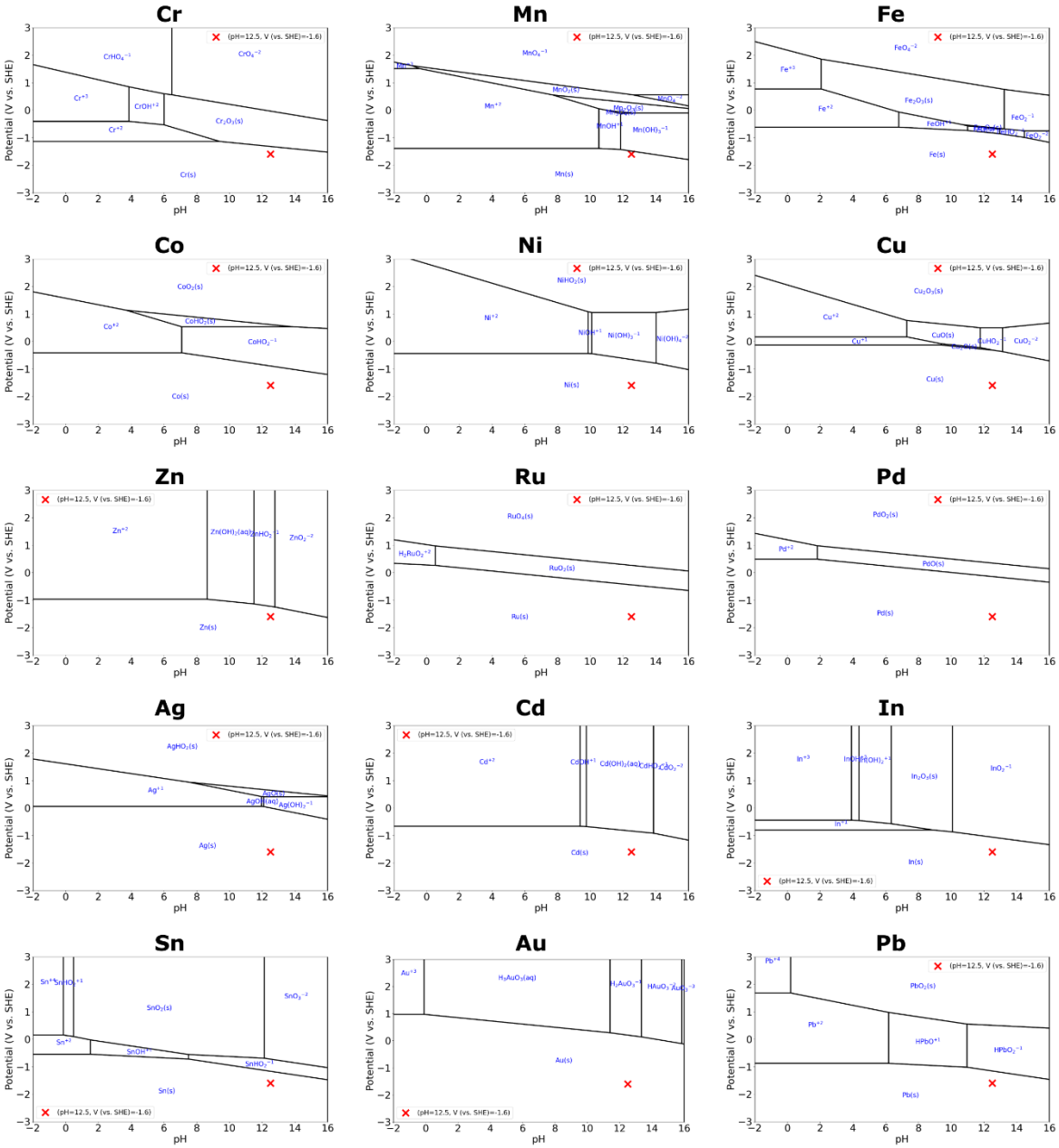

Figure S1. Pourbaix diagrams of candidate elements for CO_2R. Pourbaix diagrams showing the thermodynamically stable phases of the 15 elements considered in the compositional search space under CO_2R conditions. The red '×' marks the estimated local CO_2R conditions in 0.1 M $KHCO_3$ electrolyte, corresponding to a local pH of 12.5 and a cathode potential below −1.6 V_{SHE}.

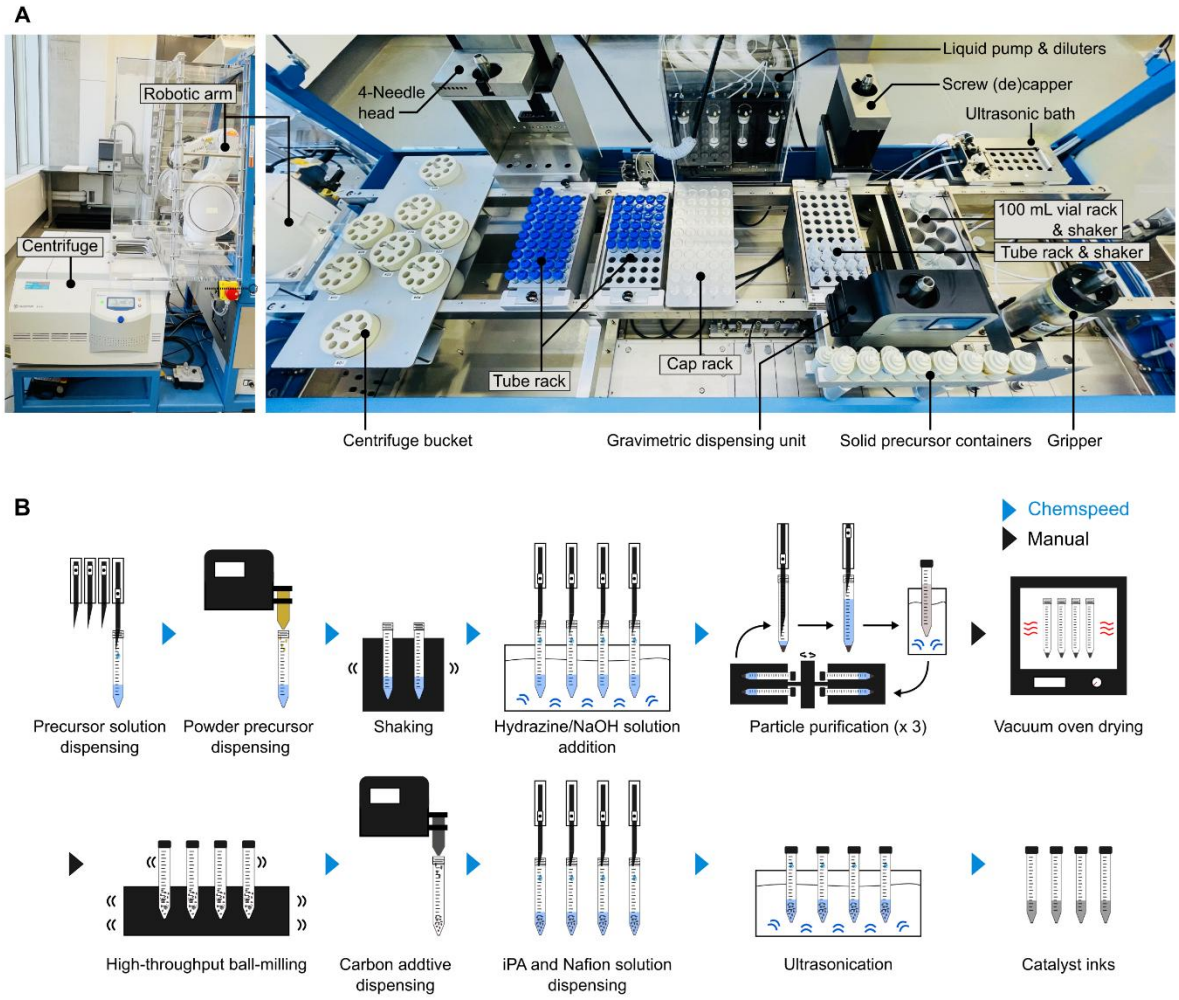

Figure S2. Automated high-throughput synthesis platform. (A) Photograph of the *Chemspeed Swing XL* platform showing the spatial configuration of robotic tools and functional components. (B) Schematic workflow of the automated high-throughput synthesis and catalyst ink preparation process for multi-element heterogeneous electrocatalysts.

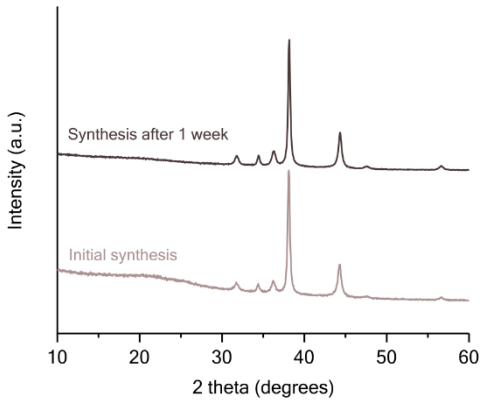

Figure S3. Repeatability of automated catalyst synthesis demonstrated by XRD patterns. XRD patterns of $Zn_{0.50}Ag_{0.50}$ catalyst synthesized at different times showing consistent crystalline structure and phase composition.

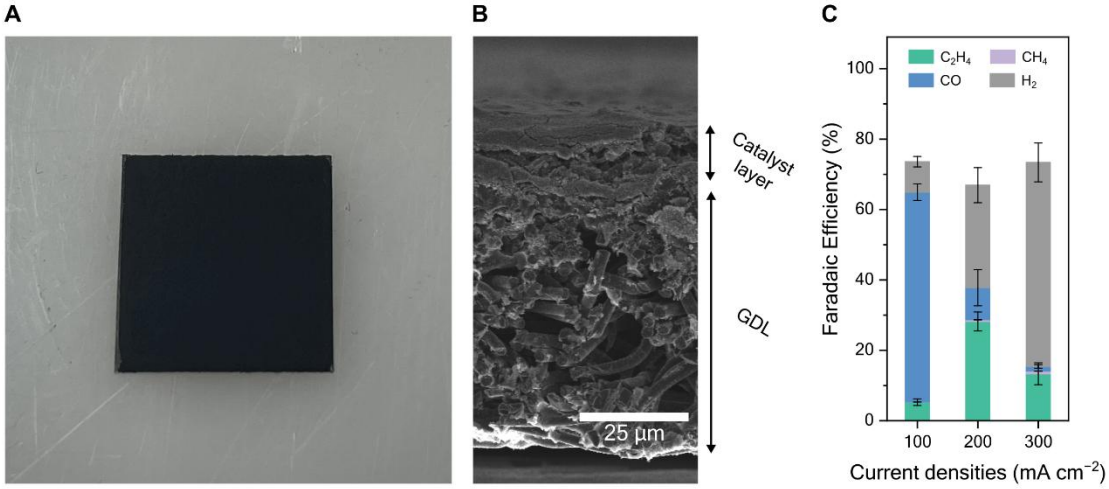

Figure S4. Characterization of drop-cast electrodes and their CO2R performance validation. (A) Digital photograph of drop-cast catalyst layer on 1 cm^2 gas diffusion layer. (B) Cross-sectional SEM image showing uniform catalyst layer formation on GDL. (C) CO_2R performance of drop-cast electrodes using commercial Cu nanoparticles (*Sigma Aldrich*) at different current densities in 0.1 M $KHCO_3$. Error bars represent the standard deviation from three independent measurements.

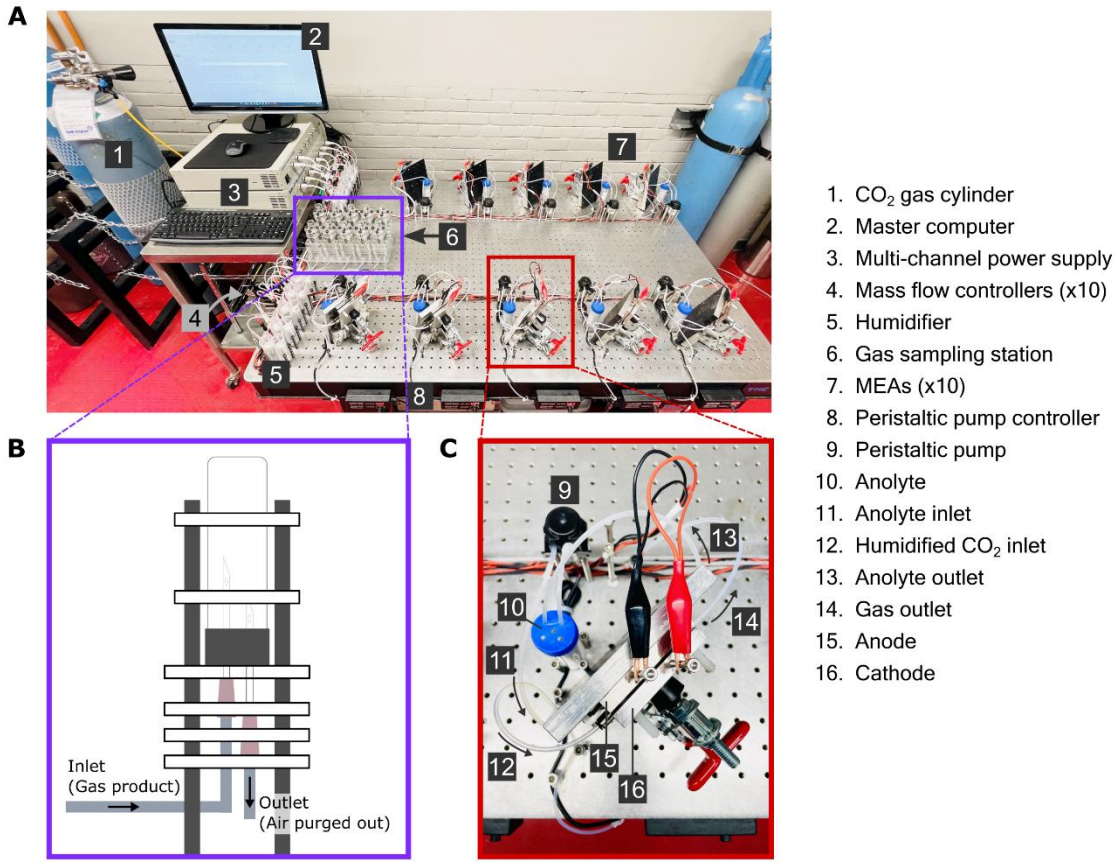

Figure S5. Membrane electrode assembly (MEA)-based parallelized electrocatalyst screening platform. (A) Overview photograph of parallelized high-throughput CO_2 MEA electrolyzers with numbered components (1–16) detailed in the legend. (B) Schematic illustration of the 10× gas sampling station. (C) Closed-up photograph of a single CO_2 MEA electrolyzer unit with fluidic connections and electrical contacts.

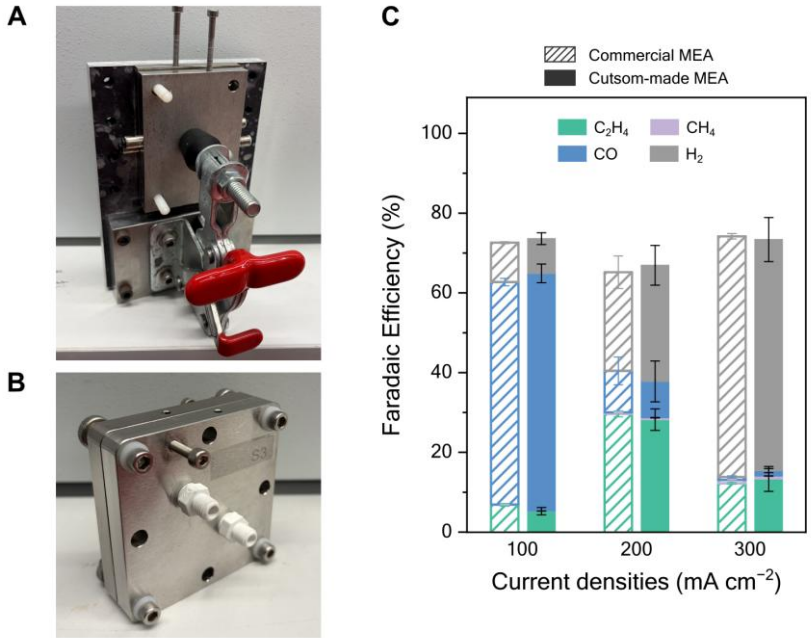

Figure S6. Comparison of conventional (commercial) MEA and custom-made MEA. (A–B) Digital photographs of custom-designed MEA cell featuring a hold-down clamp for simplified and rapid assembly[17] (A) and conventional (commercial) MEA cell (B). (C) CO_2R performance comparison between custom-made MEA and conventional MEA at different current densities. Both configurations utilized electrodes prepared by drop-casting commercial Cu nanoparticles (*Sigma-Aldrich*) and were tested in 0.1 M $KHCO_3$. Error bars represent the standard deviation from three independent measurements.

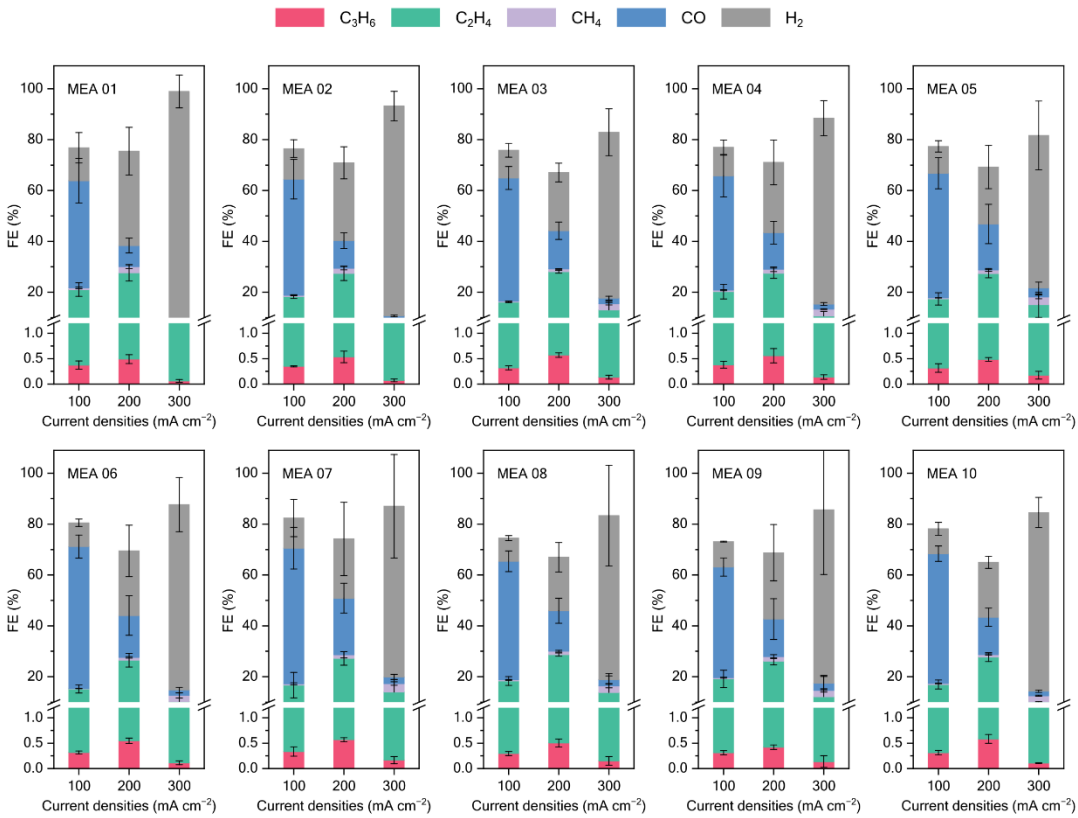

Figure S7. Validation of uniform CO_2R performance across parallel MEA cells. FE distribution for gas products at three different current densities across all ten parallel MEA cells using sputtered Cu catalyst with 750 nm thickness. Error bars represent the standard deviation from three independent measurements.

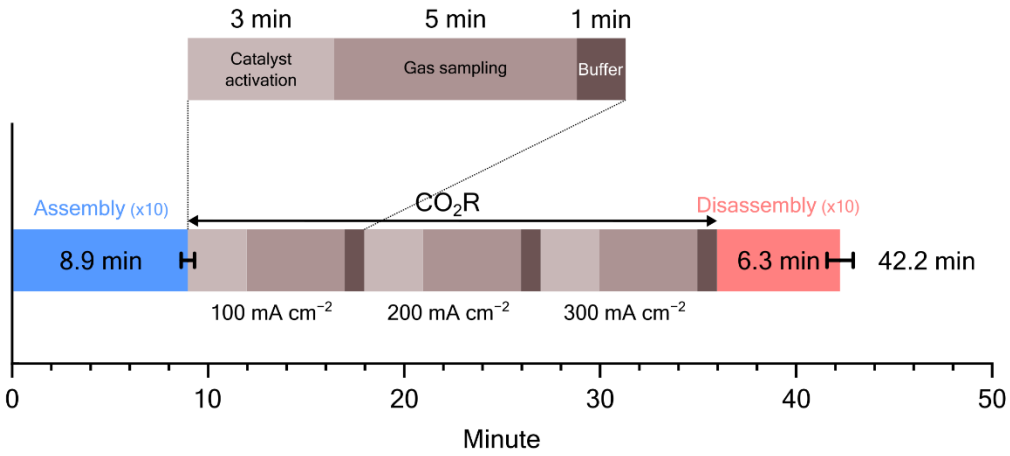

Figure S8. Time requirements for parallelized MEA system operation. Measured time profile for 10× parallel MEA system operation (42.2 minutes per round). The operation cycle consists of parallel assembly of 10 MEAs, sequential CO_2R evaluation at three current densities and parallel disassembly. Each current density measurement includes catalyst activation (3 minutes), gas sampling (5 minutes) and buffer period (1 minute). Three such rounds enable evaluation of 30 catalysts in approximately 2 hours. Error bars represent standard deviations measured from three independent operation rounds.

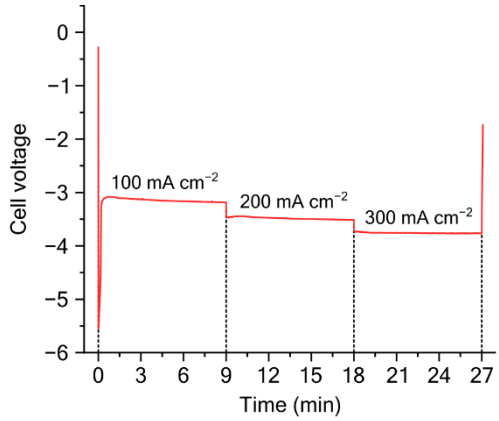

Figure S9. Voltage profile during parallel CO_2R operation at different current densities. Representative voltage profile of a $Cu_{1.00}$ catalyst operated at three different current densities (100, 200 and 300 mA cm^{-2}). Each current density region consists of a 9-minute operational period, comprising 3 minutes of catalyst activation, 5 minutes of gas sampling and a 1-minute buffer period, resulting in a total measurement time of 27 minutes.

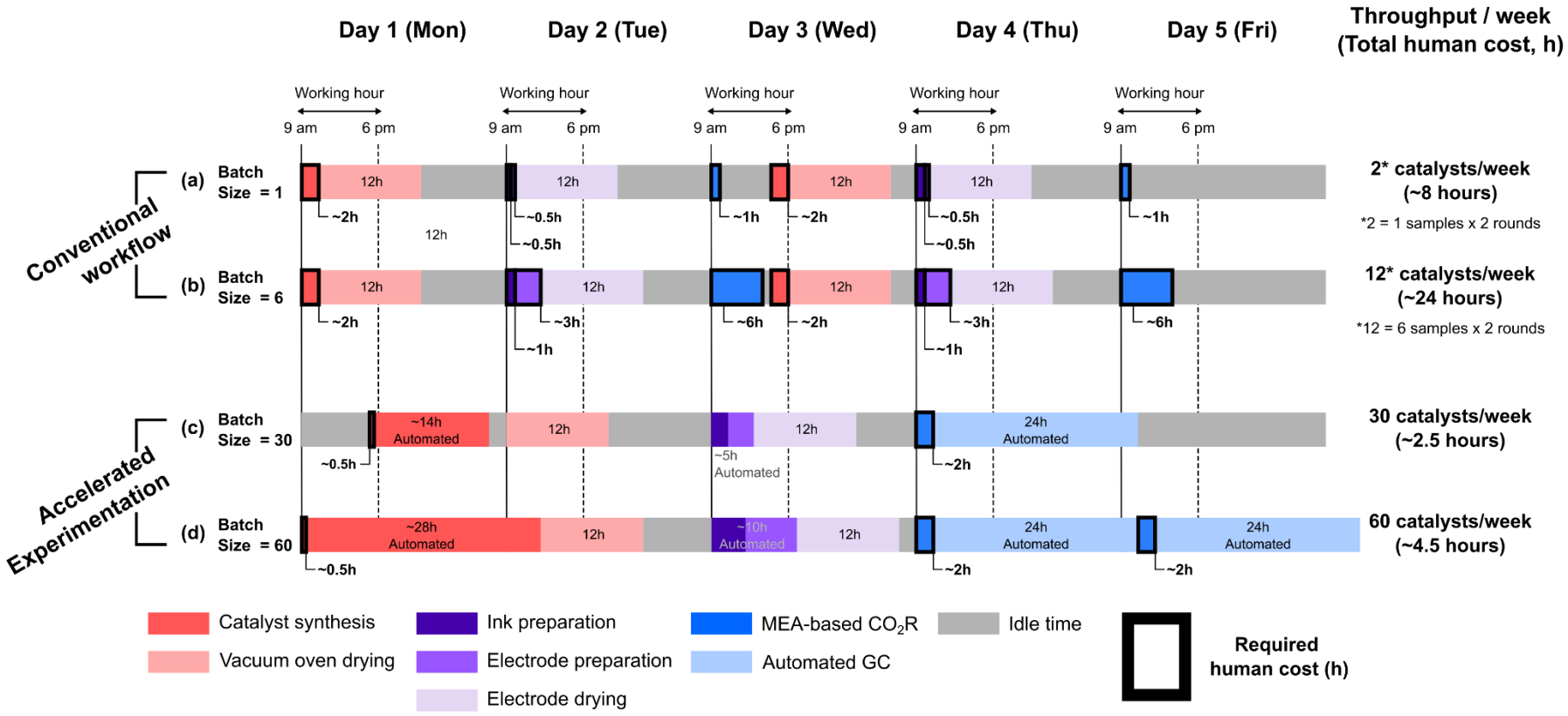

Figure S11. Weekly throughput and human time-cost comparison between conventional and accelerated experimentation workflows. Detailed timeline visualization comparing conventional and accelerated experimentation workflows over a five-day work week. **a-b**, Conventional workflow with batch sizes of 1 (**a**) and 6 (**b**). **c-d**, Accelerated experimentation workflow with batch sizes of 30 (**c**) and 60 (**d**).

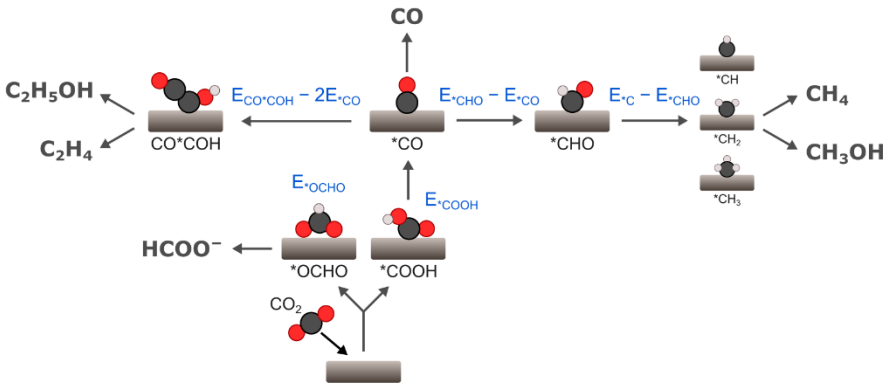

Figure S12. Possible reaction pathways for the electrocatalytic CO_2 reduction to C_{1-2} products[18-20] and five descriptors derived through feature engineering based on adsorption energies of key intermediates.

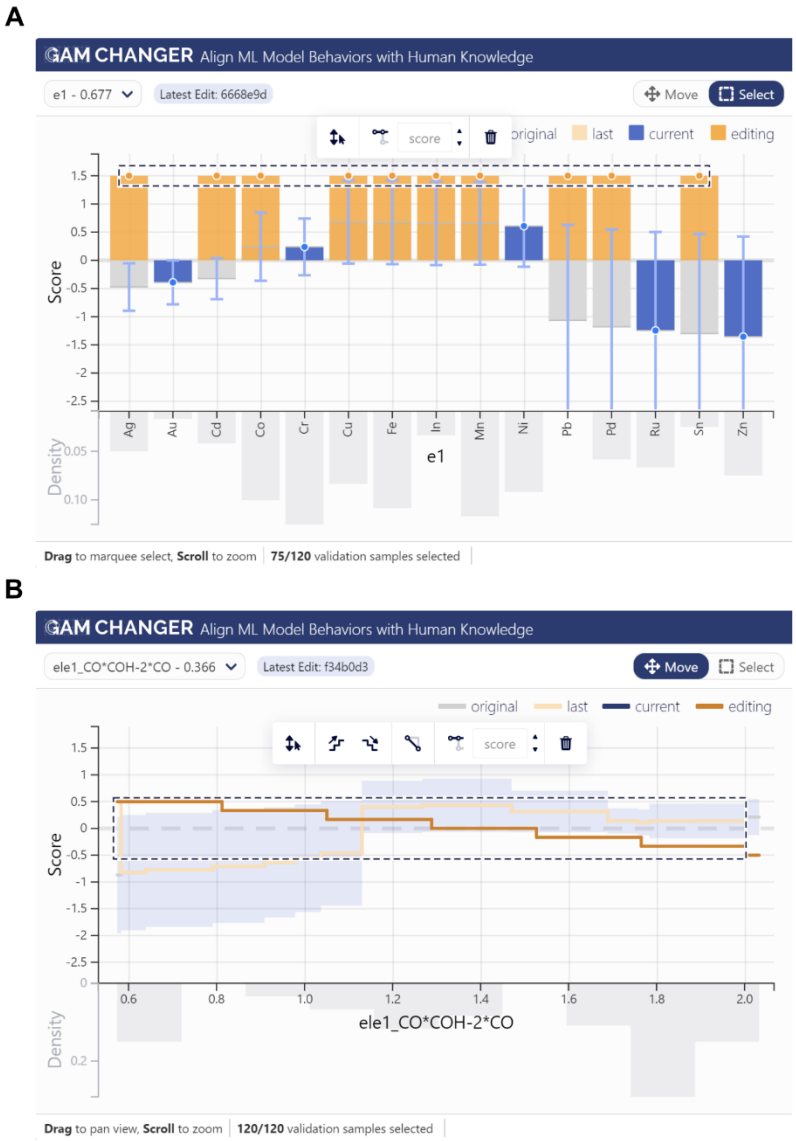

Figure S13. *GAM Changer* graphical user interface (GUI). (A–B) GAM Changer for editing shape functions of categorical (elements in the catalyst composition) (A) and E_{ads}-based continuous (B) descriptors.

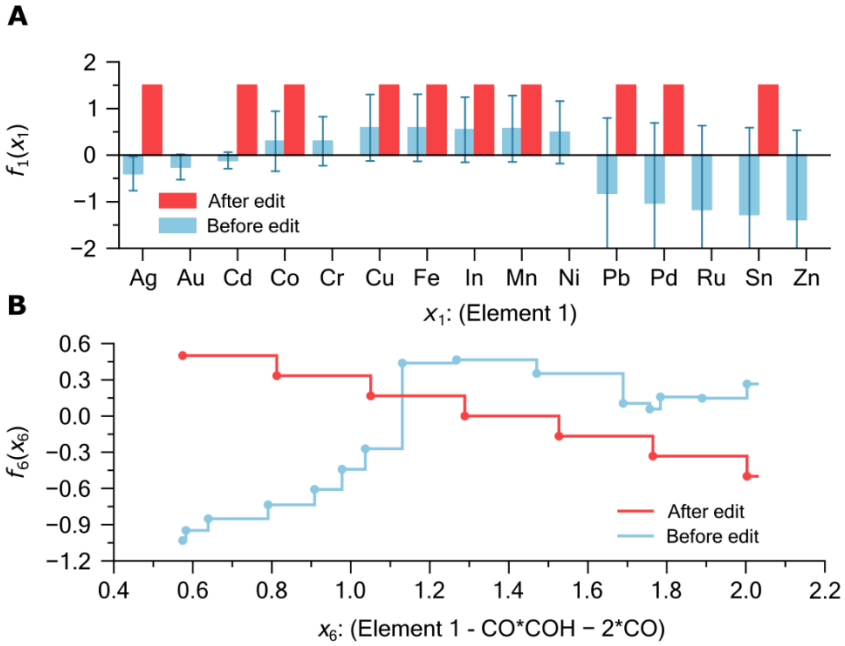

Figure S14. Example of domain knowledge-driven shape function modification of the ML model using the GAM changer. (A–B) Adjustments for a categorical element descriptor (A) and an E_{ads}-based continuous descriptor (B).

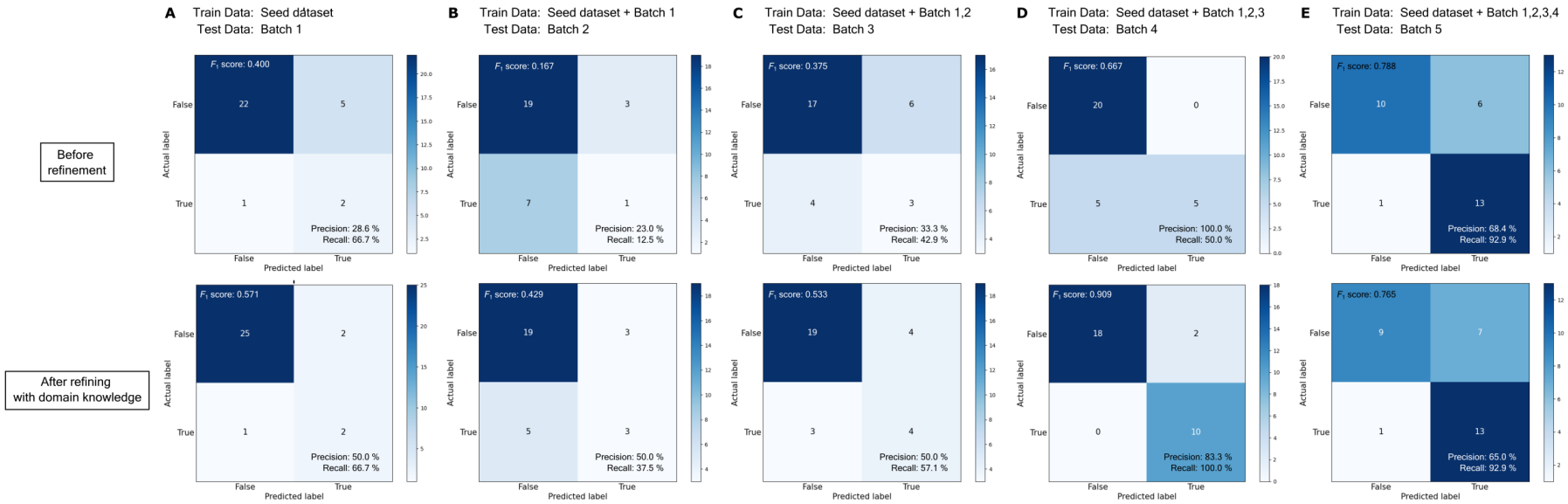

Figure S15. Confusion matrices comparing model performance before and after domain knowledge-based refinement across sequential experimental iterations.

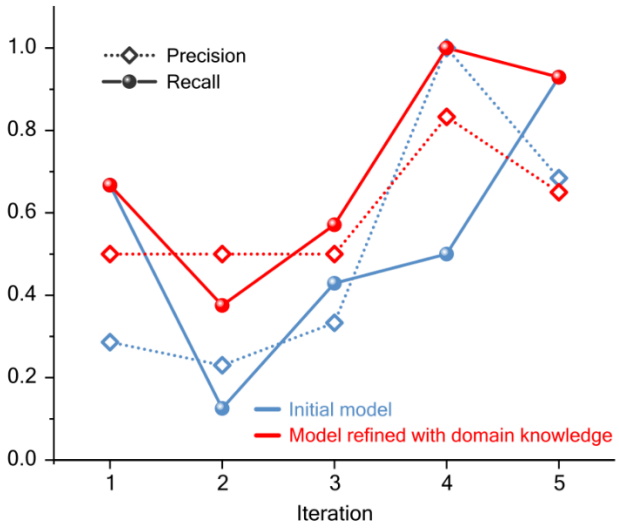

Figure S16. Evolution of precision and recall across iterations for the initial model and the domain-knowledge–refined model.

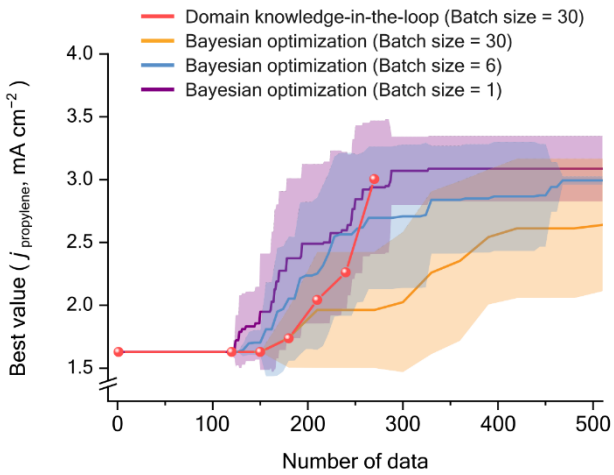

Figure S17. Comparison of different exploration strategies (including BO with batch size = 1) on the virtual composition–$j_{propylene}$ space derived from a surrogate model trained on the entire experimental dataset.

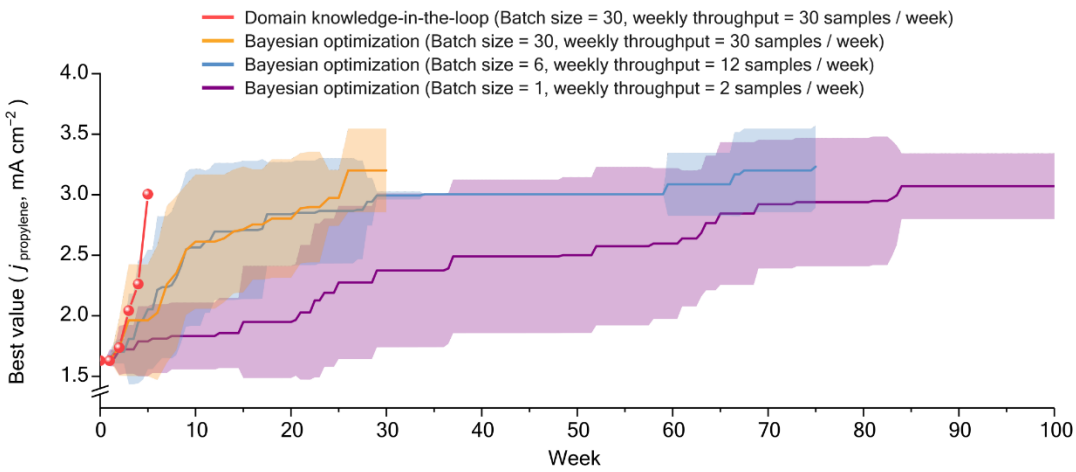

Figure S18. Time-based comparison of different exploration strategies for catalyst discovery considering realistic laboratory throughput constraints.

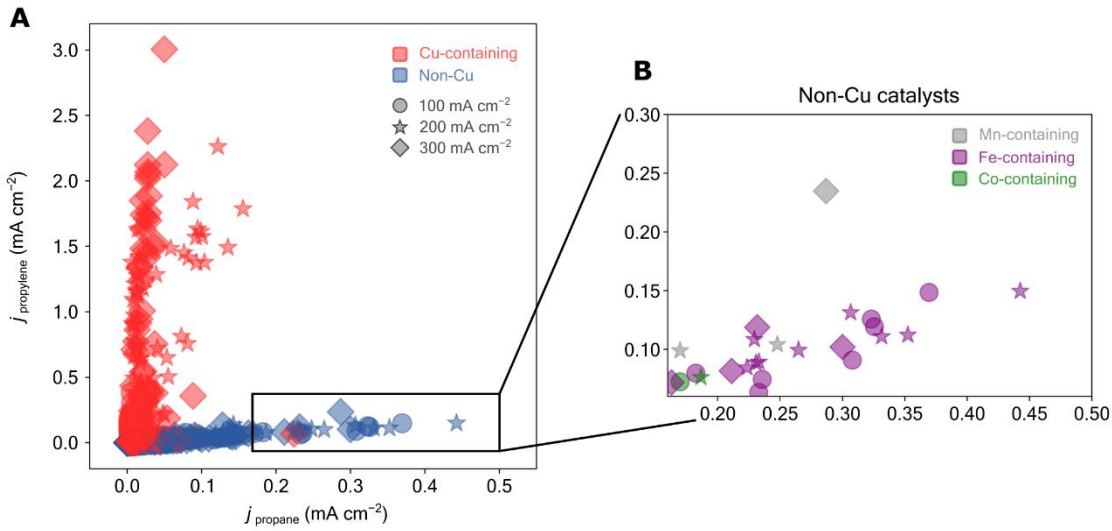

Figure S19. Distribution of propylene and propane partial current densities across the entire 300-composition dataset. (A) Scatter plot showing $j_{propane}$ versus $j_{propylene}$ for all compositions categorized by Cu content and measured at different current densities. (B) Magnified view of non-Cu catalysts with notable propane activity, highlighting contributions from Mn-, Fe-, and Co-containing compositions.

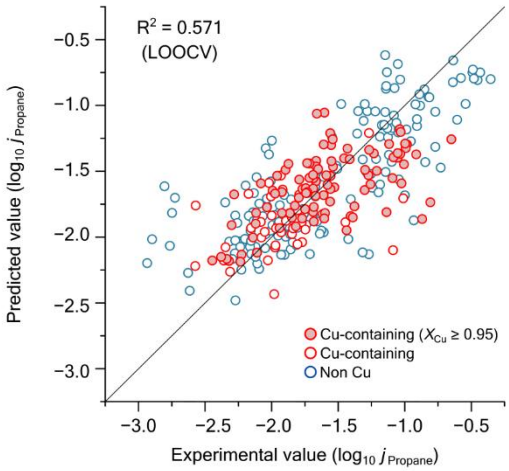

Figure S21. Correlation between experimentally observed and predicted $\log_{10} j_{propane}$ values for the entire dataset using XGBoost model, showing $R^2 = 0.571$, evaluated using leave-one-out cross-validation (LOOCV).

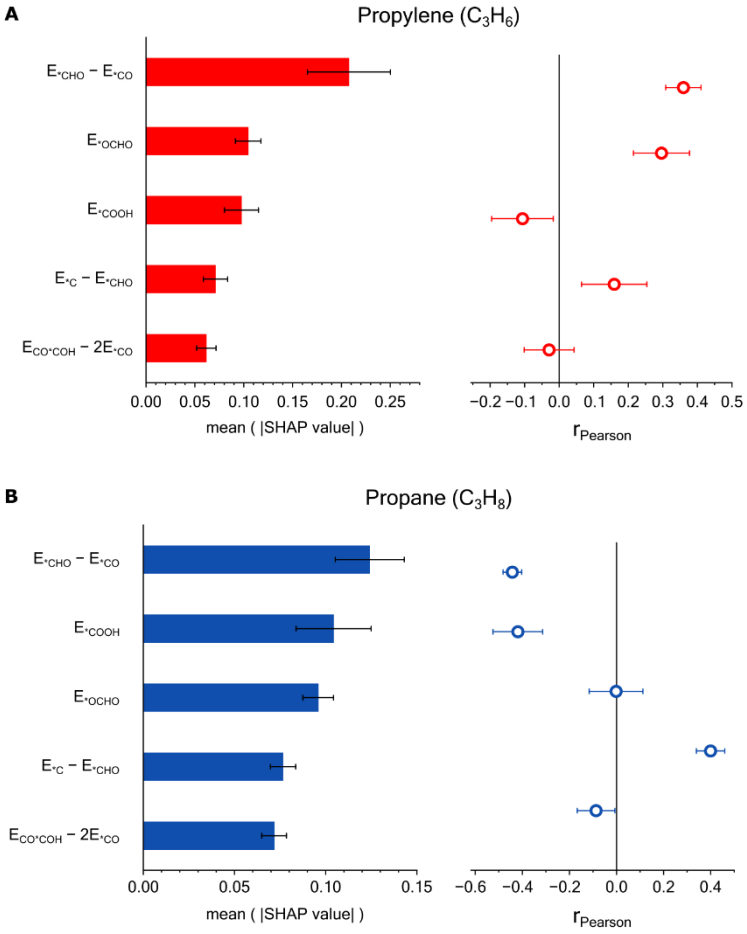

Figure S22. Consistency of feature importance and feature–SHAP correlations across ten independent training runs for propylene (A) and propane (B).

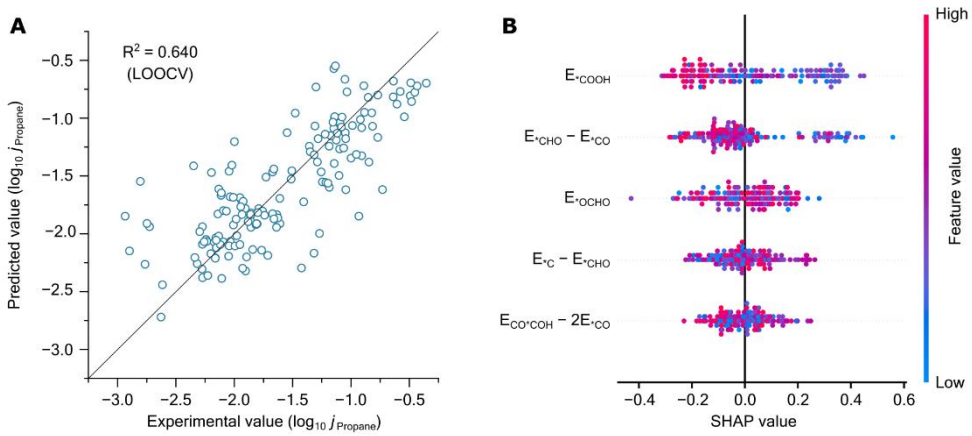

Figure S23. XGBoost model performance and feature importance analysis for propane production using only non-Cu catalyst data. (A) Correlation between experimentally observed and predicted log10 *j*propane values for non-Cu catalysts (R^2 = 0.640) using leave-one-out cross-validation (LOOCV). (B) SHAP feature importance analysis revealing the relative contribution of different E_{ads}-based descriptors to propane production in non-Cu catalysts.

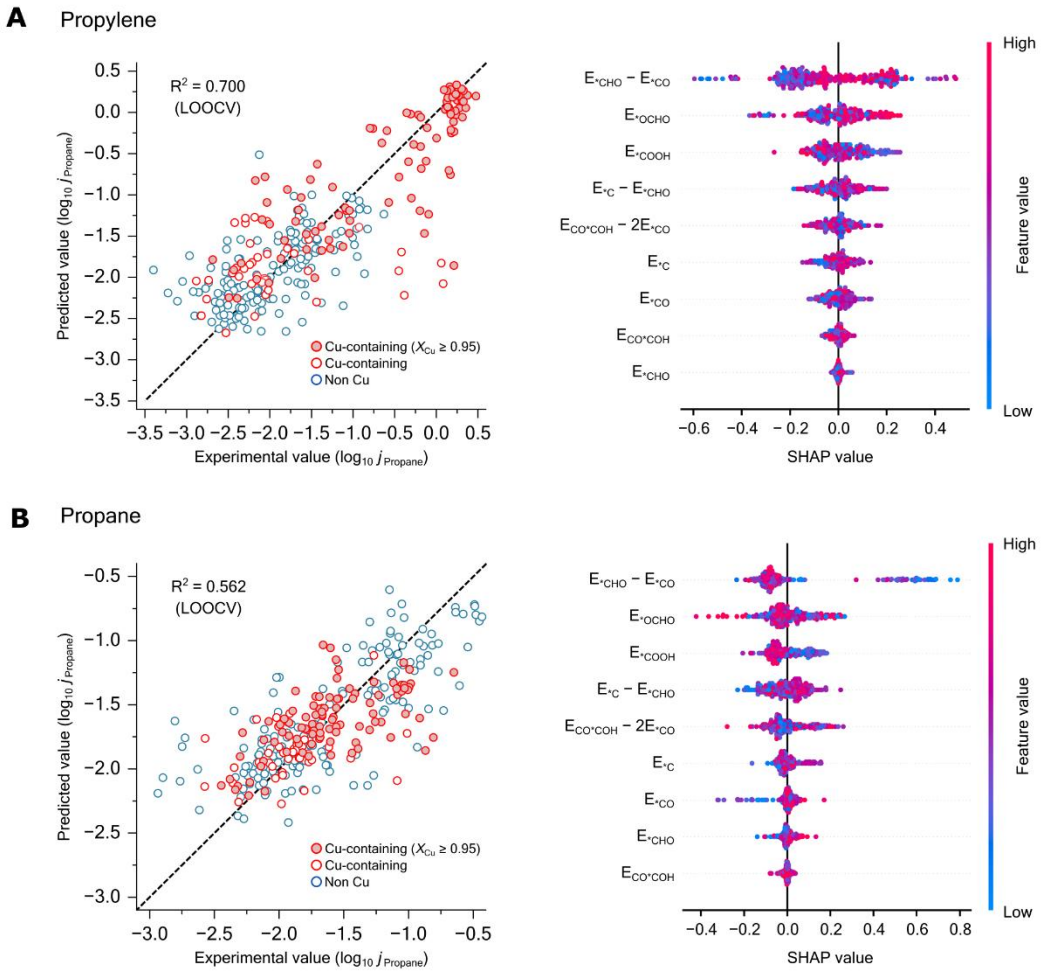

Figure S24. Comprehensive XGBoost model performance and expanded SHAP feature importance analysis for C_3 hydrocarbon production. (A–B) Model prediction performance (left) and SHAP feature importance analysis (right) including individual E_{ads} for propylene (A) and propane (B) production.

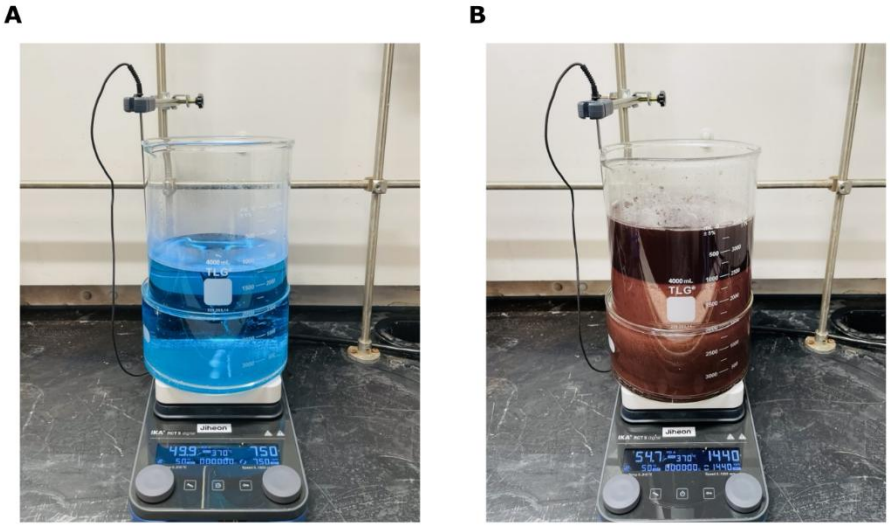

Figure S25. Scale-up synthesis of $Cu_{0.98}In_{0.02}$ catalyst. (A) Precursor solution containing Cu and In salts before addition of reducing agent. (B) Solution after addition of reducing agent showing the formation of $Cu_{0.98}In_{0.02}$ nanoparticles at 1,500 times the original synthesis scale.

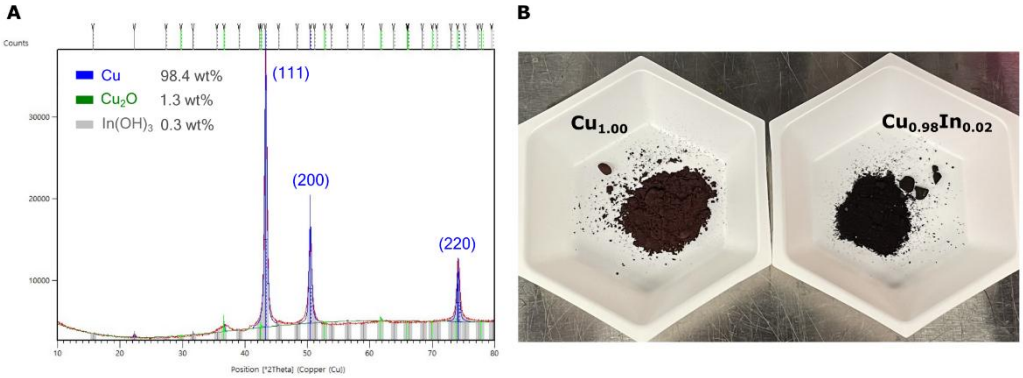

Figure S26. Phase analysis and appearance of the $Cu_{0.98}In_{0.02}$ catalyst. (A) High-resolution XRD with Rietveld quantification: Cu 98.4 wt%, Cu_2O 1.3 wt%, $In(OH)_3$ 0.3 wt%. The XRD pattern of $Cu_{0.98}In_{0.02}$ shows the allowed fcc Cu reflections at (111), (200), and (220). The (200) reflection corresponds to the {100} family in the fcc lattice, as odd-index reflections such as (100) are systematically extinct. Likewise, (211) reflections are forbidden for fcc Cu and are therefore absent in the powder XRD. The (100) and (211) facets mentioned in this work refer to surface orientations adopted for catalytic modeling (see **Method S4**) rather than observable Bragg reflections. (B) Digital photograph of $Cu_{1.00}$ (left) and $Cu_{0.98}In_{0.02}$ (right) showing a color contrast.

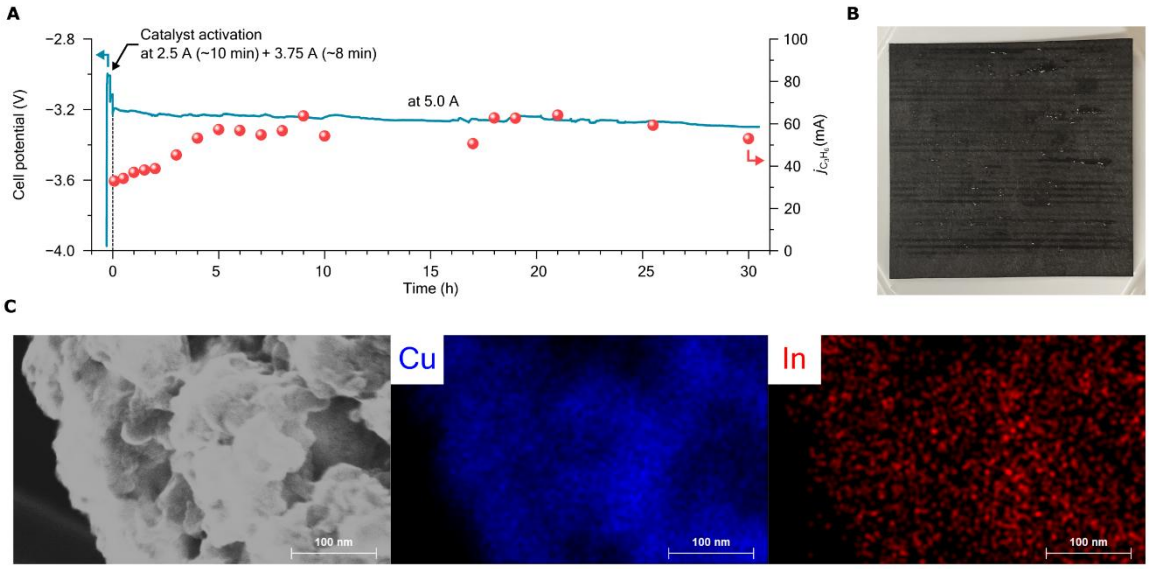

Figure S27. Device-scale stability and post-mortem analysis of the $Cu_{0.98}In_{0.02}$ catalyst. (A) Cell potential (blue, left axis) and propylene partial current j_{C3H6} (red, right axis) during 30 h operation of a 25 cm^2 MEA. (B) Back side of the cathode GDL after 30 h showing flooding streaks, indicating the need for device-level water/gas-management engineering. (C) Post-mortem TEM image and EDX elemental maps.

Tables

Table S1. Elements selected for multi-element CO_2 electrocatalyst design space and their literature precedents

Element	Literature references
Cr	21
Mn	22
Fe	23
Co	24
Ni	9,25
Cu	26
Zn	27
Ru	28
Pd	10
Ag	29
Cd	30
In	25,31
Sn	32
Au	33
Pb	15

Table S2. Detailed acc(X)eleration performance indicators (XPIs) in evaluating the acceleration performance of experimental platforms

Acc(X)eleration Performance Indicators (XPIs)	(1) Catalyst synthesis		(2) CO_2R performance evaluation		End-to-end experimentation from (1) catalyst synthesis to (2) CO_2R performance evaluation	
	Conventional lab workflow	**Accelerated experimentation**	Conventional lab workflow	**Accelerated experimentation**	Conventional lab workflow	**Accelerated experimentation**
XPI-1 Number of materials produced or examined	~6[a]	~51.4[a]	~6[a]	30[a]	~12[b]	~60[b]
XPI-2 Number of new materials with threshold performance	N/A[c]		N/A[c]		N/A[c]	
XPI-3 Performance of best material over time (fold increment)	N/A[c]		N/A[c]		N/A[c]	
XPI-4 Repeatability of the experimental platform (precision, %)	94.7%	94.5%	81.6%	82.3%	77.3%[c]	77.8%[d]
XPI-5[e] Human time-cost to achieve XPI-1 (percentage relative to conventional Edisonian-like workflow)	100%	~ 25%[f]	100%	~ 33.3%[f]	100%	18.8%[f]
Acceleration score (AS)	~5.7	~194.3 (**34.1**×)	~4.1	~ 74.1 (**18.1**×)	~9.3	~248.2 (**26.7**×)

[a] Practically achievable daily throughput (catalysts synthesized or CO_2R performance evaluated per day).
[b] Practically achievable weekly throughput (catalysts examined per week).
[c] Calculation is not applicable without a database. Note that XPI-2 and XPI-3 were not considered in the acceleration performance calculation of the experimental platforms.
[d] Overall repeatability is calculated from repeatability of (1) catalyst synthesis and (2) CO_2R performance evaluation (see **Method S2**).
[e] Detailed human time-cost to achieve the throughput is illustrated in **Figure S10**.
[f] Relative human time-cost compared to conventional lab workflow (set at 100%) required to achieve the noted throughput.

Table S3. Computational workflow for operando-consistent slab construction and adsorption-energy descriptors (AdsorbML–EquiformerV2)

	Step	Tool used	Purpose
1	Thermodynamic filtering of candidate phases	pymatgen analysis.pourbaix_diagram	Retains phases that stay metallic during CO_2R, ensuring surfaces probed by AdsorbML are in operando
2	Slab construction	pymatgen core.surface (100) & (211)	Provides two representative slabs per phase
3	Enumeration of initial adsorbate placements	AdsorbML placement module – random & heuristic placements of CO_2R intermediates	Samples binding motifs before relaxation
4	Fast geometry relaxation	EquiformerV2 (within AdsorbML) – 300 steps, fmax = 0.02 eV $Å^{-1}$	Finds low-energy structures 100x faster than DFT
5	Physical validity checks	AdsorbML filters – desorption, dissociation, slab mismatch	Removes unphysical relaxations
7	Descriptor assembly	Average E_ads across surfaces + stoichiometry features	Creates interpretable feature vector for GAM classifier

Table S4. Detailed feature edits across experimental iterations

Training Data	Target Iteration	$j_{propylene}$ Threshold [mA cm^{-2}]	Feature Type	Feature Name	Shape Function Edit Details
Seed Dataset	1	0.05	Categorical	Ele1	Set scores of ["Ag", "Cd", "Co", "Cu", "Fe", "In", "Mn", Pd", "Sn"] to 1.5.
			Categorical	Ele2	Set scores of ["Ag", "Cd", "Co", "Cu", "Fe", "In", "Mn", Pd", "Sn"] to 1.5.
			Continuous	Ele1_$E_{*CHO} - E_{*CO}$	Set 1 bins (-0.15 <= x < -0.12) to score 0.5. Set 1 bins (0.68 <= x) to score -0.5. Interpolated 15 bins (-0.15 <= x) into 5 equal-size bins.
			Continuous	Ele1_$E_{CO*COH} - 2E_{*CO}$	Set 1 bins (0.57 <= x < 0.58) to score 0.5. Set 1 bins (2 <= x) to score -0.5. Interpolated 15 bins (0.57 <= x) into 5 equal-size bins.
			Continuous	Ele2_$E_{*CHO} - E_{*CO}$	Set 1 bins (-0.15 <= x < -0.12) to score 0.5. Set 1 bins (0.68 <= x) to score -0.5. Interpolated 15 bins (-0.15 <= x) into 5 equal-size bins.
			Continuous	Ele2_$E_{CO*COH} - 2E_{*CO}$	Set 1 bins (0 <= x < 0.29) to score 0.5. Set 1 bins (2 <= x) to score -0.5. Interpolated 15 bins (0 <= x) into 5 equal-size bins.
Seed Dataset + Iteration 1	2	0.1	Categorical	Ele1	Set scores of ["Ag", "Cd", "Cu", "Fe", "In", "Ni", "Pd", "Sn"] to 1.
			Categorical	Ele2	Set scores of ["Ag", "Cd", "Cu", "Fe", "In", "Ni", "Pd", "Sn"] to 1.

Seed Dataset + Iteration 1, 2	3	0.2	Categorical	Ele1	Set scores of ["Ag", "Cd", "Co", "Cu", "Mn", "Ni", "Pd", "Pb"] to 1.
			Categorical	Ele2	Set scores of ["Ag", "Cd", "Co", "Cu", "Mn", "Ni", "Pd", "Pb"] to 1.
			Continuous	Ele1_$E_{*CHO} - E_{*CO}$	Set 1 bins (-0.15 <= x < -0.12) to score -0.5. Set 1 bins (0.68 <= x) to score 0.5. Interpolated 15 bins (-0.15 <= x) into 5 equal-size bins.
			Continuous	Ele1_$E_{CO*COH} - 2E_{*CO}$	Set 1 bins (0.57 <= x < 0.58) to score 0.5. Set 1 bins (2 <= x) to score -0.5. Interpolated 15 bins (0.57 <= x) into 5 equal-size bins.
			Continuous	Ele2_$E_{*CHO} - E_{*CO}$	Set 1 bins (-0.15 <= x < -0.12) to score -0.5. Set 1 bins (0.68 <= x) to score 0.5. Interpolated 15 bins (-0.15 <= x) into 5 equal-size bins.
			Continuous	Ele2_$E_{CO*COH} - 2E_{*CO}$	Set 1 bins (0 <= x < 0.29) to score 0.5. Set 1 bins (2 <= x) to score -0.5. Interpolated 15 bins (0 <= x) into 5 equal-size bins.
Seed Dataset + Iteration 1, 2, 3	4	0.5	Categorical	Ele1	Set scores of ["Ag", "Cd", "Cu", "Fe", "In", "Pd"] to 2.
			Categorical	Ele2	Set scores of ["Ag", "Cd", "Cu", "Fe", "In", "Pd"] to 2.
			Categorical	Ele3	Set scores of ["Ag", "Cd", "Cu", "In", "Pd"] to 2.
			Continuous	Ele3_$E_{*CHO} - E_{*CO}$	Set 1 bins (-0.08 <= x < -0.04) to score -0.5. Set 1 bins (0.68 <= x) to score 0.5. Interpolated 8 bins (-0.08 <= x) into 5 equal-size bins.
			Continuous	Ele3_$E_{CO*COH} - 2E_{*CO}$	Set 1 bins (0 <= x < 0.3) to score 0.5. Set 1 bins (1.64 <= x) to score -0.5. Interpolated 8 bins (0 <= x) into 5 equal-size bins.
Seed Dataset + Iteration 1, 2, 3, 4	5	1.0	Categorical	Ele1	Set scores of ["Ag", "Cd", "Cu", "Fe", "In", "Pd"] to 2.
			Categorical	Ele2	Set scores of ["Ag", "Cd", "Cu", "Fe", "In", "Pd"] to 2.
			Categorical	Ele3	Set scores of ["Ag", "Cd", "Cu", "In", "Pd"] to 2.

Seed Dataset + Iteration 1, 2, 3, 4, 5	6	1.0	Categorical	Ele1	Set scores of ["Ag", "Cd", "Cu", “Fe”, "In", "Pd"] to 2.
			Categorical	Ele2	Set scores of ["Ag", "Cd", "Cu", “Fe”, "In", "Pd"] to 2.
			Categorical	Ele3	Set scores of ["Ag", "Cd", "Cu", "In", "Pd"] to 2.

Table S5. Detailed acc(X)eleration performance indicators (XPIs) in evaluating the acceleration performance of different exploration strategies (see **Method S6** for detailed calculation methods)

Acc(X)eleration Performance Indicators (XPIs)					**This work (Accelerated exploration)**
	ML methodology	Bayesian optimization	Bayesian optimization	Bayesian optimization	Domain knowledge-integrated ML
	Experimental workflow (Batch size)	Conventional experimentation (Batch size = 1)	Conventional experimentation (Batch size = 6)	Accelerated experimentation (Batch size = 30)	Accelerated experimentation (Batch size = 30)
XPI-1 Number of materials produced examined (per week)		2	12	30	30
XPI-2 Number of new materials with threshold performance		N/A	N/A	N/A	N/A
XPI-3 Performance of best material over time (fold increment)		1.89	1.84	1.97	1.84
XPI-4 Repeatability of the experimental platform (precision, %)		77.3%	77.3%	77.8%	77.8%
XPI-5[e] Human time-cost to achieve XPI-1 (percentage relative to conventional Edisonian-like workflow)		100% (84 weeks × 8 hours/week = 672 hours)	121.4% (34 weeks × 24 hours/week = 816 hours)	9.7% (26 weeks × 2.5 hours/week = 65 hours)	1.9% (5 weeks × 2.5 hours/week = 12.5 hours)
Acceleration score (AS)		2.9	14.1	474.0	2314.0
Estimated discovery timeline (after seed campaign, see **Figure S10**)		~84 weeks	~34 weeks	~26 weeks	~5 weeks

Table S6. Comparison of propylene electrosynthesis performance between $Cu_{0.98}In_{0.02}$ catalyst in this work and previous literatures

Electrolyzer	Gas feed	Catalyst	Electrolyte	Propylene FE [%]	Voltage [*vs*. RHE or full-cell voltage]	Partial current density [mA cm^{-2}]	Reference
H-cell	CO_2	Cu_2O-Cu	0.1 M KCl	0.9	−1.8 V *vs*. RHE	~0.22 [a]	34
Flow cell (A = 0.72 cm^2)	CO_2	CuO-derived Cu	1.0 M KOH	0.61	−0.58 V *vs*. RHE	2.55	35
Flow cell (A = 0.33 cm^2)	CO_2:CO (8:2)	CuNC	1.0 M KOH + 0.2 M CsI	~ 1.0 [a]	−0.65 V *vs*. RHE	5.5	36
MEA (A = 25 cm^2)	CO_2	$Cu_{0.98}In_{0.02}$	0.1 M $KHCO_3$	2.0	−3.47 V	6.0	**This work**

[a] The number is calculated or estimated from the data or figure in the corresponding literature.

Table S7. Precursors for the automated multi-element electrocatalyst preparation

Element	Chemical	Company
Cr	Chromium(III) nitrate nonahydrate	*Sigma Aldrich*
Mn	Manganese(II) nitrate tetrahydrate	*Sigma Aldrich*
Fe	Iron(III) nitrate nonahydrate	*Sigma Aldrich*
Co	Cobalt(II) nitrate hexahydrate	*Sigma Aldrich*
Ni	Nickel(II) nitrate hexahydrate	*Sigma Aldrich*
Cu	Copper(II) nitrate trihydrate	*Sigma Aldrich*
Zn	Zinc(II) nitrate hexahydrate	*Sigma Aldrich*
Ru	Ruthenium(III) chloride hydrate	*Sigma Aldrich*
Pd	Palladium(II) nitrate hydrate	*Sigma Aldrich*
Ag	Silver nitrate	*Sigma Aldrich*
Cd	Cadmium chloride hydrate	*Sigma Aldrich*
In	Indium(III) nitrate hydrate	*Thermo Scientific*
Sn	Tin(IV) chloride pentahydrate	*Sigma Aldrich*
Au	Gold(III) chloride trihydrate	*Sigma Aldrich*
Pb	Lead(II) nitrate	*Sigma Aldrich*

References

1. Lan, J., Palizhati, A., Shuaibi, M., Wood, B.M., Wander, B., Das, A., Uyttendaele, M., Zitnick, C.L., and Ulissi, Z.W. (2023). AdsorbML: a leap in efficiency for adsorption energy calculations using generalizable machine learning potentials. Npj Comput. Mater. *9*, 172. https://doi.org/10.1038/s41524-023-01121-5.
2. Liao, Y.-L., Wood, B., Das, A., and Smidt, T. (2023). EquiformerV2: Improved Equivariant Transformer for Scaling to Higher-Degree Representations. Preprint at arXiv. https://doi.org/10.48550/arXiv.2306.12059.
3. Chanussot, L., Das, A., Goyal, S., Lavril, T., Shuaibi, M., Riviere, M., Tran, K., Heras-Domingo, J., Ho, C., Hu, W.H., et al. (2021). Open Catalyst 2020 (OC20) Dataset and Community Challenges. ACS Catal. *11*, 6059–6072. https://doi.org/10.1021/acscatal.0c04525.
4. Tran, R., Lan, J., Shuaibi, M., Wood, B.M., Goyal, S., Das, A., Heras-Domingo, J., Kolluru, A., Rizvi, A., Shoghi, N., et al. (2023). The Open Catalyst 2022 (OC22) Dataset and Challenges for Oxide Electrocatalysts. ACS Catal. *13*, 3066–3084. https://doi.org/10.1021/acscatal.2c05426.
5. Yao, Z.P., Lum, Y.W., Johnston, A., Mejia-Mendoza, L.M., Zhou, X., Wen, Y.G., Aspuru-Guzik, A., Sargent, E.H., and Seh, Z.W. (2023). Machine learning for a sustainable energy future. Nat. Rev. Mater. *8*, 202–215. https://doi.org/10.1038/s41578-022-00490-5.
6. Abed, J., Kim, J., Shuaibi, M., Wander, B., Duijf, B., Mahesh, S., Lee, H.S., V., G., Hoogland, S., Irtem, E., et al. (2024). Open Catalyst Experiments 2024 (OCx24): Bridging Experiments and Computational Models. Preprint at arXiv. https://https://arxiv.org/abs/2411.11783.
7. Norskov, J.K., Bligaard, T., Rossmeisl, J., and Christensen, C.H. (2009). Towards the computational design of solid catalysts. Nat. Chem. *1*, 37–46. https://doi.org/10.1038/Nchem.121.
8. Liu, X.Y., Xiao, J.P., Peng, H.J., Hong, X., Chan, K., and Norskov, J.K. (2017). Understanding trends in electrochemical carbon dioxide reduction rates. Nat. Commun. *8*, 15438. https://doi.org/10.1038/ncomms15438.
9. Zhou, Y.S., Martín, A.J., Dattila, F., Xi, S.B., López, N., Pérez-Ramírez, J., and Yeo, B.S. (2022). Long-chain hydrocarbons by CO_2 electroreduction using polarized nickel catalysts. Nat. Catal. *5*, 545–554. https://doi.org/10.1038/s41929-022-00803-5.
10. Xie, Y., Ou, P.F., Wang, X., Xu, Z.Y., Li, Y.C., Wang, Z.Y., Huang, J.E., Wicks, J., McCallum, C., Wang, N., et al. (2022). High carbon utilization in CO_2 reduction to multi-carbon products in acidic media. Nat. Catal. *5*, 564–570. https://doi.org/10.1038/s41929-022-00788-1.
11. Gajdos, M., Eichler, A., Hafner, J., Meyer, G., and Rieder, K.H. (2005). CO adsorption on a Cu(211) surface: First-principle calculation and STM study. Phys. Rev. B *71*, 035402. https://doi.org/10.1103/PhysRevB.71.035402.
12. Papanikolaou, K.G., Darby, M.T., and Stamatakis, M. (2019). CO-Induced Aggregation and Segregation of Highly Dilute Alloys: A Density Functional Theory Study. J. Phys. Chem. C *123*, 9128–9138. https://doi.org/10.1021/acs.jpcc.9b00649.
13. Toyao, T., Suzuki, K., Kikuchi, S., Takakusagi, S., Shimizu, K., and Takigawa, I. (2018). Toward Effective Utilization of Methane: Machine Learning Prediction of Adsorption Energies on Metal Alloys. J. Phys. Chem. C *122*, 8315–8326. https://doi.org/10.1021/acs.jpcc.7b12670.
14. Grote, J.P., Zeradjanin, A.R., Cherevko, S., Savan, A., Breitbach, B., Ludwig, A., and

Mayrhofer, K.J.J. (2016). Screening of material libraries for electrochemical CO_2 reduction catalysts - Improving selectivity of Cu by mixing with Co. J. Catal. *343*, 248–256. https://doi.org/10.1016/j.jcat.2016.02.026.

15. Fang, W.S., Guo, W., Lu, R.H., Yan, Y., Liu, X.K., Wu, D., Li, F.M., Zhou, Y.S., He, C.H., Xia, C.F., et al. (2024). Durable CO_2 conversion in the proton-exchange membrane system. Nature *626*, 86–91. https://doi.org/10.1038/s41586-023-06917-5.
16. Bahari, Z., Dichi, E., Legendre, B., and Dugué, J. (2003). The equilibrium phase diagram of the copper-indium system:: a new investigation. Thermochim. Acta. *401*, 131–138. https://doi.org/10.1016/S0040-6031(02)00500-2.
17. Kose, T., O'Brien, C.P., Wicks, J., Abed, J., Xiao, Y.C., Sutherland, B., Sarkar, A., Jaffer, S.A., Sargent, E.H., and Sinton, D. (2022). High-throughput parallelized testing of membrane electrode assemblies for CO_2 reduction. Catal. Sci. Technol. *12*, 6239–6245. https://doi.org/10.1039/d2cy00873d.
18. Kortlever, R., Shen, J., Schouten, K.J.P., Calle-Vallejo, F., and Koper, M.T.M. (2015). Catalysts and Reaction Pathways for the Electrochemical Reduction of Carbon Dioxide. J. Phys. Chem. Lett. *6*, 4073–4082. https://doi.org/10.1021/acs.jpclett.5b01559.
19. Feaster, J.T., Shi, C., Cave, E.R., Hatsukade, T.T., Abram, D.N., Kuhl, K.P., Hahn, C., Norskov, J.K., and Jaramillo, T.F. (2017). Understanding Selectivity for the Electrochemical Reduction of Carbon Dioxide to Formic Acid and Carbon Monoxide on Metal Electrodes. ACS Catal. *7*, 4822–4827. https://doi.org/10.1021/acscatal.7b00687.
20. Todorova, T.K., Schreiber, M.W., and Fontecave, M. (2020). Mechanistic Understanding of CO_2 Reduction Reaction (CO2RR) Toward Multicarbon Products by Heterogeneous Copper-Based Catalysts. ACS Catal. *10*, 1754–1768. https://doi.org/10.1021/acscatal.9b04746.
21. Moreno, J.J., Hooe, S.L., and Machan, C.W. (2021). DFT Study on the Electrocatalytic Reduction of CO_2 to CO by a Molecular Chromium Complex. Inorg. Chem. *60*, 3635–3650. https://doi.org/10.1021/acs.inorgchem.0c03136.
22. Zhang, B.X., Zhang, J.L., Shi, J.B., Tan, D.X., Liu, L.F., Zhang, F.Y., Lu, C., Su, Z.Z., Tan, X.N., Cheng, X.Y., et al. (2019). Manganese acting as a high-performance heterogeneous electrocatalyst in carbon dioxide reduction. Nat. Commun. *10*, 2980. https://doi.org/10.1038/s41467-019-10854-1.
23. Zhou, B.W., Ou, P.F., Pant, N., Cheng, S.B., Vanka, S., Chu, S., Rashid, R.T., Botton, G., Song, J., and Mi, Z.T. (2020). Highly efficient binary copper-iron catalyst for photoelectrochemical carbon dioxide reduction toward methane. Proc. Natl. Acad. Sci. U.S.A. *117*, 1330–1338. https://doi.org/10.1073/pnas.1911159117.
24. Preikschas, P., Zhang, J., Seemakurthi, R.R., Lian, Z., Martín, A.J., Xi, S.B., Krumeich, F., Ma, H.B., Zhou, Y.S., López, N., et al. (2024). CO_2 Electroreduction to Long-Chain Hydrocarbons on Cobalt Catalysts. Adv. Energy. Mater. *14*, 2401447. https://doi.org/10.1002/aenm.202401447.
25. Fan, Z.Z., Luo, R.C., Zhang, Y.X., Zhang, B., Zhai, P.L., Zhang, Y.T., Wang, C., Gao, J.F., Zhou, W., Sun, L.C., and Hou, J.G. (2023). Oxygen-Bridged Indium-Nickel Atomic Pair as Dual-Metal Active Sites Enabling Synergistic Electrocatalytic CO_2 Reduction. Angew. Chem. Int. Ed. *62*, e202216326. https://doi.org/10.1002/anie.202216326.
26. Hori, Y., Wakebe, H., Tsukamoto, T., and Koga, O. (1994). Electrocatalytic Process of CO Selectivity in Electrochemical Reduction of CO_2 at Metal-Electrodes in Aqueous-Media. Electrochim. Acta *39*, 1833-1839. https://doi.org/10.1016/0013-4686(94)85172-7.

27. Wan, L.L., Zhang, X.L., Cheng, J.S., Chen, R., Wu, L.X., Shi, J.W., and Luo, J.S. (2022). Bimetallic Cu-Zn Catalysts for Electrochemical CO_2 Reduction: Phase-Separated versus Core-Shell Distribution. ACS Catal. *12*, 2741–2748. https://doi.org/10.1021/acscatal.1c05272.
28. Wang, X., Ou, P.F., Ozden, A., Hung, S.F., Tam, J., Gabardo, C.M., Howe, J.Y., Sisler, J., Bertens, K., de Arquer, F.P.G., et al. (2022). Efficient electrosynthesis of *n*-propanol from carbon monoxide using a Ag-Ru-Cu catalyst. Nat. Energy *7*, 170–176. https://doi.org/10.1038/s41560-021-00967-7.
29. Frisch, M.L., Wu, L.F., Atlan, C., Ren, Z., Han, M., Tucoulou, R., Liang, L., Lu, J.S., Guo, A., Nong, H.N., et al. (2023). Unraveling the synergistic effects of Cu-Ag tandem catalysts during electrochemical CO_2 reduction using nanofocused X-ray probes. Nat. Commun. *14*, 7833. https://doi.org/10.1038/s41467-023-43693-2.
30. Maeng, J.Y., Hwang, S.Y., Kim, Y.J., Yoon, I., Myung, C.W., Rhee, C.K., and Sohn, Y. (2023). Electrochemical CO_2/CO Reduction and C-C Coupling Path for Mimicking Fischer-Tropsch Synthesis over Cadmium Electrodes. J. Phys. Chem. C *127*, 11448–11461. https://doi.org/10.1021/acs.jpcc.3c02571.
31. Cheng, Q., Huang, M., Xiao, L., Mou, S.Y., Zhao, X.L., Xie, Y.Q., Jiang, G.D., Jiang, X.Y., and Dong, F. (2023). Unraveling the Influence of Oxygen Vacancy Concentration on Electrocatalytic CO_2 Reduction to Formate over Indium Oxide Catalysts. ACS Catal. *13*, 4021–4029. https://doi.org/10.1021/acscatal.2c06228.
32. Ren, W.H., Tan, X., Qu, J.T., Li, S.S., Li, J.T., Liu, X., Ringer, S.P., Cairney, J.M., Wang, K.X., Smith, S.C., and Zhao, C. (2021). Isolated copper-tin atomic interfaces tuning electrocatalytic CO_2 conversion. Nat. Commun. *12*, 1449. https://doi.org/10.1038/s41467-021-21750-y.
33. Chen, Y.H., Li, C.W., and Kanan, M.W. (2012). Aqueous CO_2 Reduction at Very Low Overpotential on Oxide-Derived Au Nanoparticles. J. Am. Chem. Soc. *134*, 19969–19972. https://doi.org/10.1021/ja309317u.
34. Lee, S., Kim, D., and Lee, J. (2015). Electrocatalytic Production of C3-C4 Compounds by Conversion of CO_2 on a Chloride-Induced Bi-Phasic Cu_2O-Cu Catalyst. Angew. Chem. Int. Ed. *54*, 14701–14705. https://doi.org/10.1002/anie.201505730.
35. Ting, L.R.L., García-Muelas, R., Martín, A.J., Veenstra, F.L.P., Chen, S.T.J., Peng, Y.J., Per, E.Y.X., Pablo-García, S., López, N., Pérez-Ramírez, J., and Yeo, B.S. (2020). Electrochemical Reduction of Carbon Dioxide to 1-Butanol on Oxide-Derived Copper. Angew. Chem. Int. Ed. *59*, 21072–21079. https://doi.org/10.1002/anie.202008289.
36. Gao, J., Bahmanpour, A., Kröcher, O., Zakeeruddin, S.M., Ren, D., and Grätzel, M. (2023). Electrochemical synthesis of propylene from carbon dioxide on copper nanocrystals. Nat. Chem. *15*, 705–713. https://doi.org/10.1038/s41557-023-01163-8.